MULTIMODAL EMOTION RECOGNITION SYSTEM

Project Report

Submitted by

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Project Repository:
https://github.com/vishakhmv/IIIT_project

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DECLARATION

I, **Vishakh M V**, Department of Computer Science and Engineering, Saintgits College of Engineering (Autonomous), Kottayam, hereby declare that the project report entitled **"MULTIMODAL EMOTION RECOGNITION"** is a record of the original work carried out by me.

The methodologies, experimental results, and analyses presented in this report are based on my own implementation and study.

Place: Kottayam **Vishakh M V**

Date: 24/05/2026

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Vishakh M V

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Abbreviations

AI	–	Artificial Intelligence
BERT	–	Bidirectional Encoder Representations from Transformers
BiLSTM	–	Bidirectional Long Short-Term Memory
CSV	–	Comma-Separated Values
GPU	–	Graphics Processing Unit
HuBERT	–	Hidden-Unit BERT
ID	–	Identification
MFCC	–	Mel-Frequency Cepstral Coefficients
MLP	–	Multi-Layer Perceptron
NSP	–	Next Sentence Prediction
ReLU	–	Rectified Linear Unit
TESS	–	Toronto Emotional Speech Set
TF-IDF	–	Term Frequency-Inverse Document Frequency
t-SNE	–	t-Distributed Stochastic Neighbor Embedding
WAV	–	Waveform Audio File Format
CSS	–	Cascading Style Sheets
HTML	–	HyperText Markup Language
UI	–	User Interface

Abstract

Emotion recognition plays a significant role in improving human-computer interaction systems by enabling machines to understand and interpret human emotional states. This project presents a Multimodal Emotion Recognition system capable of classifying human emotions using speech-only, text-only, and combined speech-text modalities.

The proposed system was developed using the TESS (Toronto Emotional Speech Set) dataset containing seven emotional categories: Angry, Disgust, Fear, Happy, Neutral, Pleasant Surprise, and Sad. Three independent deep learning pipelines were implemented and evaluated. The Speech-Only architecture combines pretrained HuBERT contextual speech embeddings, MFCC acoustic features, and Bidirectional Long Short-Term Memory (BiLSTM) networks for temporal speech modeling. The Text-Only architecture utilizes pretrained DistilBERT transformer embeddings for semantic emotion understanding. The Multimodal Fusion architecture integrates both speech and textual representations to learn unified emotional embeddings.

The experimental results demonstrate that speech-based representations play a dominant role in emotion recognition for linguistically constrained datasets such as TESS. The Speech-Only model achieved an accuracy of 99.64%, while the Text-Only model achieved only 14.29% due to limited semantic variation within the dataset. The Multimodal Fusion model achieved 99.29% accuracy by combining complementary speech and text representations.

In addition to quantitative evaluation metrics, the project also includes confusion matrix analysis, t-SNE latent space visualization, and multimodal representation learning analysis to study model behavior and emotional feature separability. The results demonstrate the effectiveness of multimodal deep learning techniques for robust emotion recognition and highlight the importance of acoustic speech information in emotional understanding systems.

Chapter 1

Introduction

Emotion recognition is an important research area in artificial intelligence, affective computing, and human-computer interaction. It focuses on enabling machines to identify and interpret human emotional states from different modalities such as speech, text, facial expressions, and physiological signals. Emotion-aware systems can improve the quality of intelligent applications by enabling more adaptive and human-centered interactions.
Recent advances in deep learning and transformer-based architectures have significantly improved the capability of emotion recognition systems by learning high-level contextual and temporal representations directly from raw data.

1.1 Problem Statement

Human emotions are expressed through multiple modalities such as speech, language, facial expressions, and physiological signals. Accurately recognizing these emotions is important for building intelligent and emotionally aware systems. However, understanding emotions using only a single modality can lead to limited performance and poor generalization.
Speech-based emotion recognition systems capture acoustic emotional cues such as tone, pitch, rhythm, and prosody, while text-based systems focus on semantic and contextual understanding of language. Individually, each modality has its own limitations. Text-only systems may fail when emotional information is primarily conveyed through vocal delivery, whereas speech-only systems may ignore semantic context.
This project aims to address these challenges by developing and comparing three independent emotion recognition architectures:

- A Speech-Only model for learning acoustic emotional representations
- A Text-Only model for semantic emotion understanding
- A Multimodal Fusion model that combines both speech and text representations

The project further investigates how different modalities contribute to emotion recognition performance and analyzes whether multimodal fusion can improve emotional understanding compared to standalone unimodal approaches.

1.2 Objectives

The main objectives of this project are:

- To develop a Speech-Only emotion recognition model using HuBERT embeddings, MFCC features, and BiLSTM networks.
- To implement a Text-Only emotion recognition model using pretrained DistilBERT embeddings.
- To design a Multimodal Fusion architecture that combines speech and text representations for emotion classification.
- To evaluate and compare the performance of Speech-Only, Text-Only, and Fusion-based architectures.
- To analyze model performance using confusion matrices, learning curves, and t-SNE visualizations.

1.3 Significance of the Project

Emotion-aware intelligent systems have important applications in healthcare, virtual assistants, customer support, education, mental health monitoring, and human-computer interaction. Accurate emotion recognition enables machines to interact more naturally, intelligently, and empathetically with human users.

Modern conversational AI systems and chatbots can greatly benefit from emotion recognition capabilities. By understanding a user's emotional state through speech and language, intelligent assistants can generate more adaptive, context-aware, and emotionally sensitive responses. This can improve user engagement, communication quality, and overall user experience in real-world interactive systems.

This project demonstrates the effectiveness of deep learning-based multimodal approaches for emotion recognition using speech and text modalities. The study also highlights the importance of acoustic speech representations in emotionally constrained datasets such as TESS and provides practical insights into transformer-based representation learning, multimodal fusion techniques, and affective computing systems.

Chapter 2

Literature Review

Recent advancements in deep learning and transformer-based architectures have significantly improved emotion recognition systems in speech and natural language processing. Modern affective computing systems increasingly rely on contextual representation learning and multimodal fusion techniques to improve emotional understanding and classification robustness.

2.1 Speech Representation Learning using HuBERT

Hsu et al. (2021) introduced HuBERT, a self-supervised approach to speech representation learning that significantly improved contextual acoustic modeling. Traditional speech emotion recognition systems mainly relied on handcrafted acoustic features, which often failed to capture complex vocal nuances and contextual speech information.

The HuBERT architecture learns contextual speech representations directly from raw audio waveforms through masked prediction mechanisms. This approach enables the model to capture high-level acoustic and phonetic information such as tone, prosody, rhythm, and speaking style, which are highly important for emotion recognition tasks. The effectiveness of transformer-based acoustic representation learning presented in this work was also considered while developing the Speech-Only architecture implemented in this project.

Available at: https://doi.org/10.1109/TASLP.2021.3122291

2.2 Text Representation Learning using DistilBERT

Sanh et al. (2019) proposed DistilBERT, a lightweight distilled version of the BERT transformer architecture designed for efficient natural language understanding. Traditional text-based emotion recognition methods such as bag-of-words and TF-IDF representations often struggled to capture contextual semantic meaning from language data.

DistilBERT utilizes knowledge distillation techniques to preserve most of BERT's contextual language understanding capability while significantly reducing computational complexity and inference time. This makes DistilBERT highly suitable for efficient emotion recognition systems requiring contextual semantic embeddings. The contextual language modeling capability provided by DistilBERT forms the basis of the Text-Only architecture used in this project. Available at: https://doi.org/10.48550/arXiv.1910.01108

2.3 Multimodal Fusion for Emotion Recognition

Poria et al. (2017) presented a comprehensive review of affective computing techniques and emphasized the transition from unimodal analysis to multimodal fusion approaches. The study highlighted that human emotions are inherently multimodal and cannot always be accurately understood using only a single modality.

The authors demonstrated that combining speech and textual information improves emotional understanding by capturing complementary emotional cues from heterogeneous modalities. Multimodal fusion approaches provide better robustness and generalization compared to standalone unimodal systems. This research directly supports the Multimodal Fusion architecture implemented in this project, where speech and text representations are combined to improve emotion classification performance.

Available at: https://doi.org/10.1016/j.inffus.2017.02.003

2.4 Summary

The literature demonstrates that transformer-based contextual representation learning and multimodal fusion techniques provide highly effective solutions for emotion recognition tasks. The Speech-Only, Text-Only, and Multimodal Fusion architectures implemented in this project are strongly motivated by these research contributions.

Chapter 3

Dataset

The models implemented in this project are trained and evaluated using the TESS (Toronto Emotional Speech Set) dataset. The dataset is widely used for speech emotion recognition research due to its balanced emotional distribution and high-quality audio recordings. Dataset Source: https://www.kaggle.com/datasets/ejlok1/toronto-emotional-speech-set-tess

3.1 Dataset Overview

The TESS dataset contains 2,800 speech samples recorded by two female actresses aged 26 and 64. The dataset is balanced across seven emotional categories:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Pleasant Surprise
- Sad

Figure 3.1 shows the balanced emotion distribution of the TESS dataset.

3.2 Linguistic Structure

The dataset uses linguistically neutral carrier phrases such as:

> “Say the word door”
> “Say the word neat”

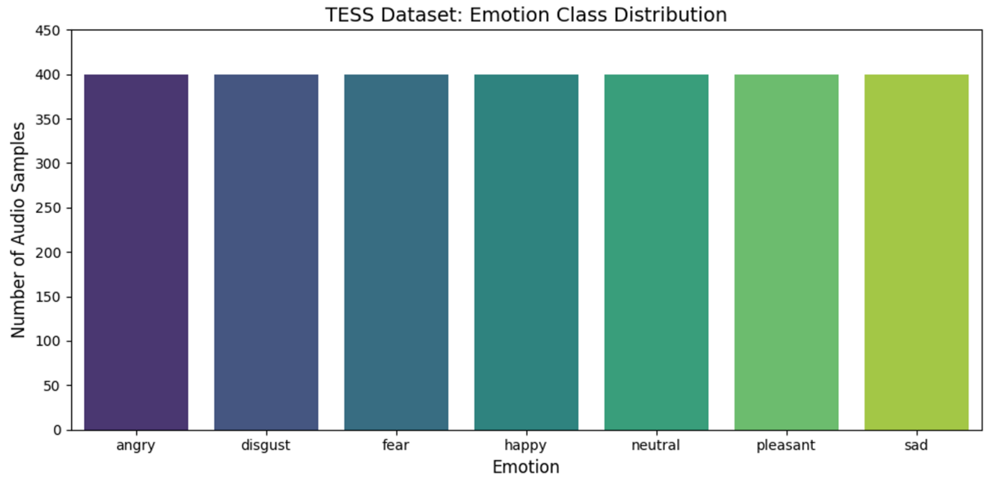


Figure 3.1: Emotion Class Distribution in the TESS Dataset

The same spoken sentences are repeated across all emotional categories, ensuring that emotional variation primarily originates from vocal expression rather than changes in textual content. This characteristic makes the dataset highly suitable for analyzing the contribution of acoustic speech features in emotion recognition systems.

3.3 Dataset Significance

The balanced structure of the TESS dataset helps reduce class imbalance issues during model training and evaluation. Additionally, the controlled linguistic content enables effective comparison between Speech-Only, Text-Only, and Multimodal Fusion architectures by isolating emotional cues present in speech and text modalities.

Chapter 4

Data Extraction and Splitting

The TESS dataset was extracted and processed using custom Python preprocessing scripts. The complete data pipeline includes dataset extraction, metadata generation, transcript reconstruction, emotion label encoding, and train-test splitting.

Figure 4.1: TESS Dataset Processing Pipeline

4.1 Data Extraction

The compressed TESS dataset archive was extracted into a local working directory using Python preprocessing utilities.

4.1.1 Extraction Pipeline

The extraction pipeline performed the following preprocessing operations:

- Initialized the computational workspace
- Loaded the compressed dataset archive
- Extracted audio files using Python `zipfile` utilities
- Verified extracted folder structure and audio contents

4.1.2 Output Structure

The extracted dataset structure is shown below:

```
/IIIT_project/TESS_dataset/
 TESS Toronto emotional speech set data/
```

The extracted audio files were verified before preprocessing to ensure correct dataset integrity and folder organization.

4.2 Data Splitting and Metadata Generation

After extraction, the complete TESS dataset was recursively processed to generate structured metadata required for multimodal training and evaluation.
The preprocessing pipeline created paired speech-text samples by extracting:

- Relative audio file paths
- Reconstructed text transcripts
- Encoded emotion labels

4.2.1 Folder Parsing

The dataset directory was recursively processed using Python's `os.walk()` utility. Folder names and audio filenames were automatically parsed to construct structured metadata stored in a Pandas DataFrame.

4.2.2 Transcript Generation

The TESS dataset does not provide separate text transcripts. Since the dataset follows fixed sentence structures, text transcripts were reconstructed directly from the audio filenames. For example, given the filename:

```
YAF_back_angry.wav
```

The filename encodes the Speaker ID, spoken word, and emotion label. The preprocessing pipeline automatically extracted the spoken word and reconstructed the transcript using the fixed carrier phrase format:

```
say the word back
```

These generated transcripts serve as semantic inputs for the Text-Only and Multimodal Fusion architectures.

4.2.3 Example Speech-Text Pairs

Audio File	**Generated Text**
YAF_back_angry.wav	say the word back
YAF_door_happy.wav	say the word door

Table 4.1: Example Speech-Text Pairs

4.2.4 Emotion Label Encoding

Each string-based emotion category was mapped to a numerical integer label for supervised model training.

Emotion	**Encoded Label**
Angry	0
Disgust	1
Fear	2
Happy	3
Neutral	4
Pleasant Surprise	5
Sad	6

Table 4.2: Emotion Label Encoding

4.2.5 Train-Test Split

The complete dataset was divided into an 80% training split and a 20% testing split. A stratified sampling technique, controlled by a fixed random seed, was utilized to preserve balanced emotion distributions across both subsets.

4.2.6 Generated CSV Files and Structure

The preprocessing pipeline finalized the data preparation by exporting the structured metadata into two CSV files:

```
IIIT_project/Data_split/
 train_split.csv
 test_split.csv
```

Each CSV file contains the following structured columns:

Column	Description
path	Relative path to the audio sample
text	Generated text transcript
label_encoded	Numerical emotion label

Table 4.3: CSV File Structure

4.2.7 Outcome

The final preprocessing pipeline successfully produced a clean, reproducible, and structured multimodal dataset suitable for Speech-Only, Text-Only, and Multimodal Fusion training and evaluation.

Chapter 5

System Architecture

This chapter describes the architectures implemented for Speech-Only, Text-Only, and Multimodal Fusion emotion recognition. The architectural design decisions were explicitly made to effectively capture emotional information from speech signals and textual representations using deep learning techniques.

5.1 Speech-Only Architecture

The Speech-Only architecture predicts emotions directly from raw speech audio by combining pretrained contextual speech embeddings, handcrafted acoustic features, and temporal sequence modelling techniques.

The architecture is designed to learn both high-level contextual speech semantics and low-level acoustic characteristics associated with emotional expression in speech signals.

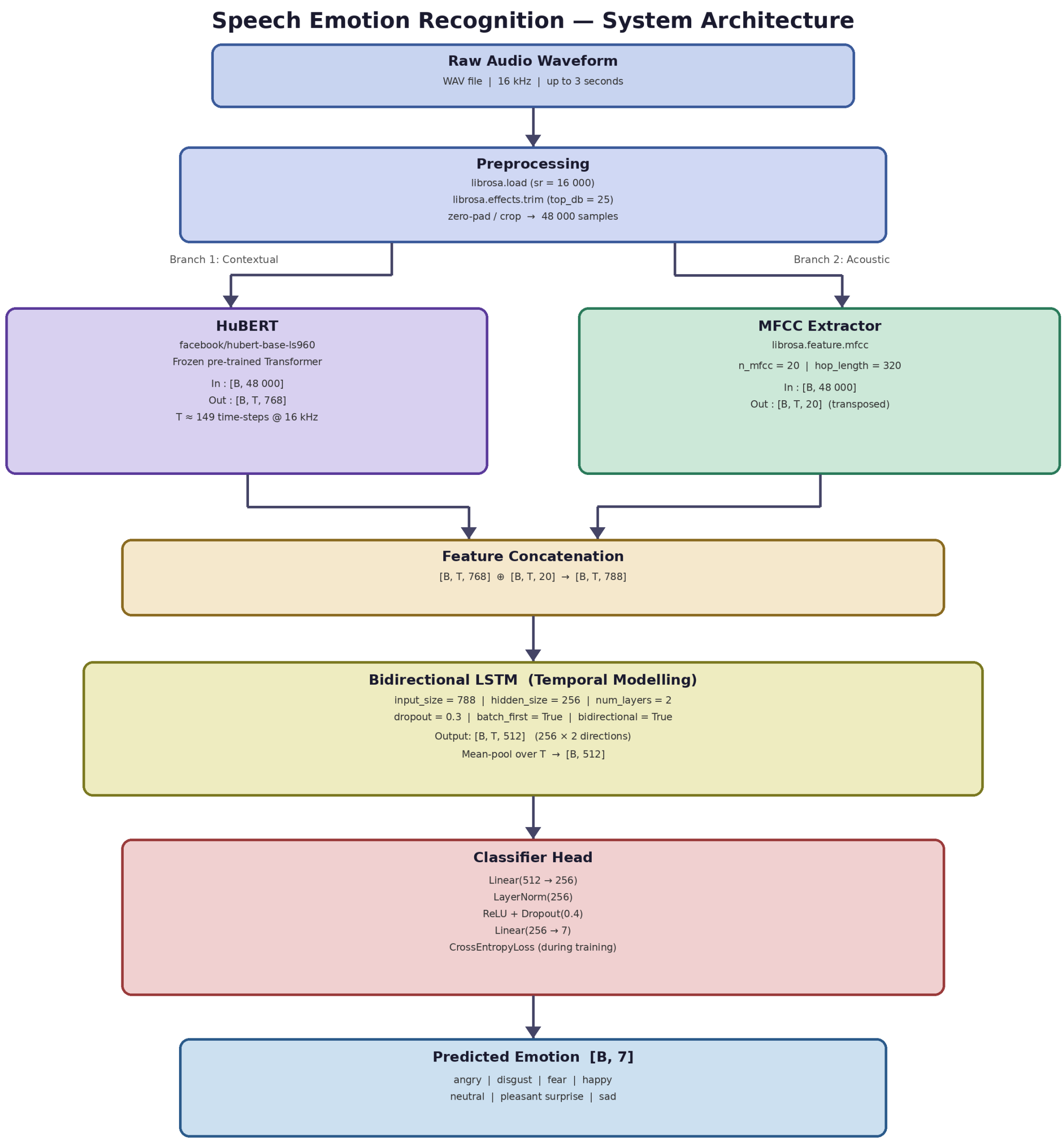


Figure 5.1: Speech-Only Emotion Recognition System Architecture

5.1.1 Preprocessing

The Speech-Only model receives raw speech audio waveforms (16 kHz) as input. Each audio sample undergoes several preprocessing operations using the `librosa` library before feature extraction.

The preprocessing pipeline includes:

- Audio loading at a 16,000 Hz sampling rate
- Silence trimming (`top_db=25`)
- Fixed-length padding and truncation to exactly 48,000 samples (3 seconds)

Design Rationale: All audio samples are standardized to a fixed duration of 3 seconds to ensure consistent tensor dimensions during neural network batch processing. Furthermore, silence trimming helps the model focus primarily on active vocal expressions rather than irrelevant silent regions, thereby improving computational efficiency and the quality of the extracted acoustic features.

5.1.2 Feature Extraction

The architecture extracts two complementary speech representations simultaneously using a dual-branch approach.

HuBERT Contextual Speech Embeddings

The primary branch passes the raw speech waveform through a pretrained HuBERT model (`facebook/hubert-base-ls960`) to generate contextual speech embeddings. This outputs a feature tensor of dimensions [Batch, Time, 768].
HuBERT captures:

- Speech context and prosodic information
- Temporal speech structure
- High-level acoustic semantics and emotional characteristics

Design Rationale: HuBERT is utilized because transformer-based contextual speech embeddings are highly effective in capturing the deep semantic characteristics present in human speech. The pretrained HuBERT weights are explicitly frozen during training to preserve these generalized speech representations and to drastically reduce computational overhead.

MFCC Acoustic Features

In parallel, the secondary branch extracts Mel-Frequency Cepstral Coefficients (MFCC) using `librosa.feature.mfcc` with `n_mfcc=20` and `hop_length=320`. The transposed output yields a feature tensor of dimensions [Batch, Time, 20].
Design Rationale: MFCC features are incorporated because they provide important, biologically-grounded low-level acoustic information (such as pitch, tone, and vocal tract variations) that may not be fully captured by contextual transformer embeddings alone.

5.1.3 Feature Fusion

The HuBERT embeddings [Batch, Time, 768] and MFCC acoustic features [Batch, Time, 20] are concatenated along the feature dimension to create a unified speech representation tensor of dimensions [Batch, Time, 788].

Design Rationale: Feature fusion is performed to combine high-level semantic information with low-level acoustic characteristics, resulting in a richer, more robust emotional representation than either independent method could provide.

5.1.4 Temporal Modelling

The concatenated speech representation is passed through a 2-layer Bidirectional Long Short-Term Memory (BiLSTM) network with a hidden size of 256. This yields a sequential output tensor of dimensions [Batch, Time, 512].
The BiLSTM models sequential emotional dependencies across speech frames and learns temporal speech dynamics including:

- Emotional transitions and speaking rhythm
- Temporal prosodic variations
- Sequential acoustic patterns

Design Rationale: A BiLSTM architecture is utilized because speech emotion recognition is inherently sequential; emotional characteristics evolve dynamically across time frames. Bidirectional processing specifically enables the network to capture contextual dependencies from both forward and backward directions, ensuring complete temporal context.

5.1.5 Temporal Pooling

After BiLSTM processing, global temporal mean pooling is applied across all time steps. This operation compresses the [Batch, Time, 512] sequential outputs into a compact, fixed-dimensional emotional embedding of [Batch, 512].
Design Rationale: Temporal mean pooling is mathematically necessary to squash the variable time dimension. It generates a stable, fixed-length summary vector representing the overall emotional characteristics of the entire 3-second speech sample, which is required for the final classification layers.

5.1.6 Classification Head

The pooled emotional embedding [Batch, 512] is passed through a fully connected classification head consisting of:

- Linear dimensionality reduction ($512 \rightarrow 256$)
- Layer normalization
- ReLU activation coupled with Dropout ($p = 0.4$)
- Final Linear classification layer ($256 \rightarrow 7$)

The classifier maps the learned emotional embedding to one of the seven emotion categories (angry, disgust, fear, happy, neutral, pleasant surprise, sad) present in the TESS dataset.
Design Rationale: Layer normalization is incorporated to improve training stability and convergence speed. Simultaneously, the 40% dropout regularization is heavily utilized to reduce overfitting, ensuring the model generalizes well to unseen speech samples rather than memorizing the training data.

5.1.7 Final Emotion Prediction

The final output of the Speech-Only architecture is a [Batch, 7] tensor representing the probability distribution across all emotion classes. The emotion with the highest probability score is selected as the predicted emotional state of the input speech sample.

5.2 Text-Only Architecture

The Text-Only architecture performs emotion classification by learning semantic and contextual language representations directly from textual transcripts. It utilizes a pretrained transformer-based language model to extract deep linguistic features.
The architecture is specifically designed to isolate and evaluate the semantic emotional content of the spoken words, completely independent of the acoustic delivery.

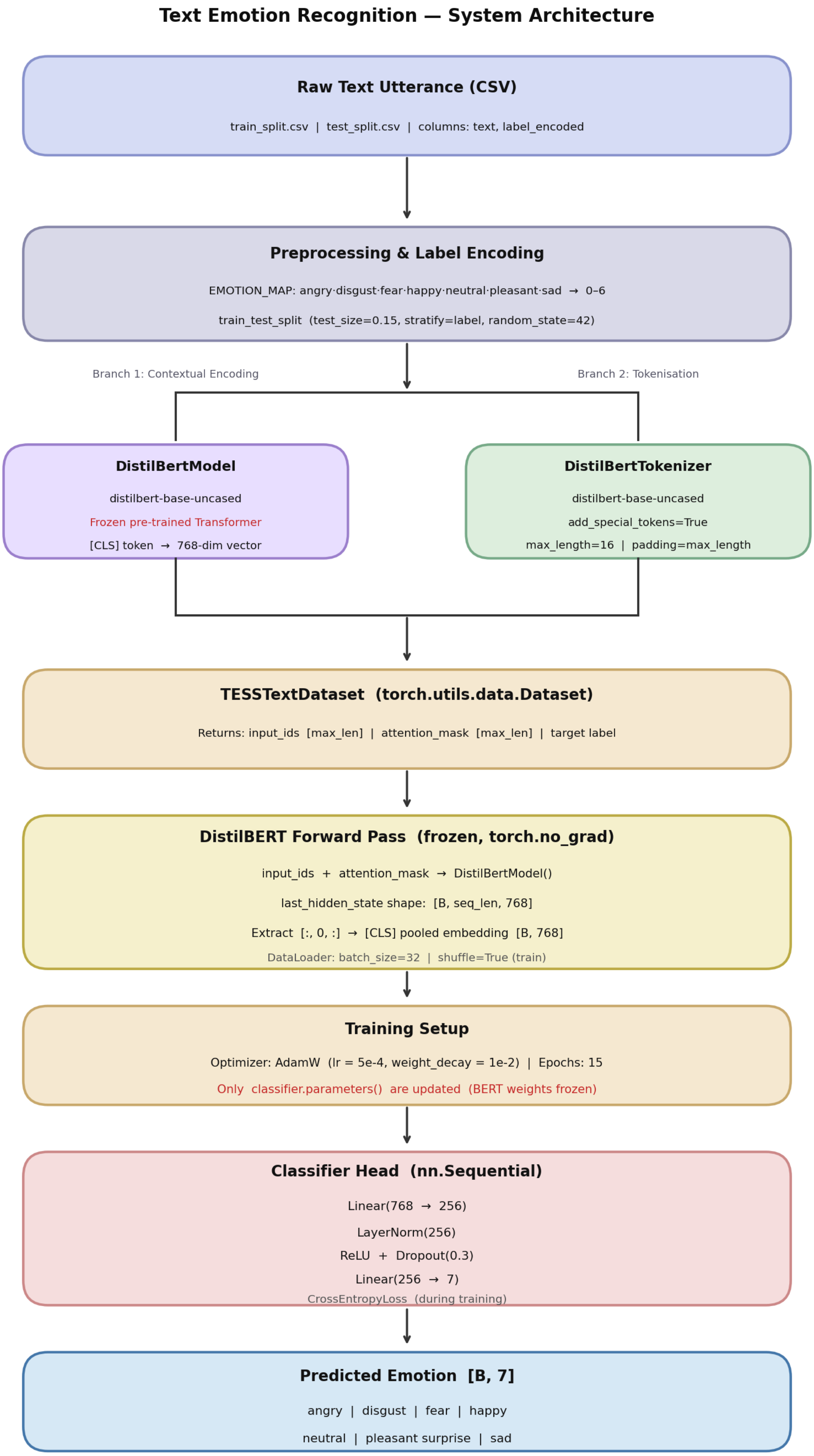


Figure 5.2: Text-Only Emotion Recognition System Architecture

5.2.1 Preprocessing and Tokenization

The model receives textual input sequences representing the speech transcripts (e.g., "say the word back"). The raw text is processed using the Hugging Face `DistilBertTokenizer` (`distilbert-base-uncased`).
The tokenizer converts the raw text into transformer-compatible representations by generating two specific outputs:

- **Input IDs:** Numerical vocabulary mappings of the text tokens.
- **Attention Masks:** Binary tensors indicating which tokens are actual words versus padding.

Special tokens (`[CLS]` and `[SEP]`) are automatically prepended and appended. The sequences are padded or truncated to a `max_length` of 16, resulting in input tensors of dimensions [Batch, 16].
Design Rationale: Converting raw text into mapped token IDs and attention masks is mathematically required for transformer encoders. A maximum sequence length of 16 was explicitly chosen because the TESS dataset relies on short, fixed carrier phrases. Enforcing this strict length ensures perfectly uniform [Batch, 16] tensors, optimizing GPU memory utilization and batch processing speeds without losing any semantic data.

5.2.2 Contextual Modelling

The tokenized text tensors (`input_ids` and `attention_mask`) are passed through a pretrained `DistilBertModel` (`distilbert-base-uncased`).
DistilBERT acts as a deep contextual encoder, generating high-dimensional embeddings that capture linguistic structure, word relationships, and sentence-level semantics. The forward pass outputs a hidden state tensor of dimensions [Batch, 16, 768].
Design Rationale: DistilBERT is selected over the standard BERT model because it retains 97% of BERT's language understanding capabilities while being 60% faster and significantly lighter. Furthermore, the pretrained transformer parameters are explicitly frozen (`requires_grad = False`) during training. This prevents the model from catastrophically forgetting its massive pretraining corpus and heavily protects against overfitting, as the TESS dataset has extremely low semantic variance.

5.2.3 CLS Sentence Representation

After transformer encoding, the architecture must extract a single vector that represents the entire sentence. The model isolates the contextual embedding corresponding to the very first token, the `[CLS]` (Classification) token, using the slicing operation `outputs.last_hidden_state[:, 0, :]`.
This operation compresses the [Batch, 16, 768] sequence tensor into a compact [Batch, 768] pooled embedding.

Design Rationale: In BERT-family architectures, the `[CLS]` token is specifically pre-trained using Next Sentence Prediction (NSP) and sequence-level tasks to act as an aggregate representation of the entire input sequence. Extracting this specific token is the most mathematically sound method for reducing a variable-token sentence into a fixed-length semantic vector for classification.

5.2.4 Classification Head

The pooled `[CLS]` embedding [Batch, 768] is passed through a fully connected classification head consisting of:

- Linear dimensionality reduction ($768 \rightarrow 256$)
- Layer normalization
- ReLU activation coupled with Dropout ($p = 0.3$)
- Final Linear classification layer ($256 \rightarrow 7$)

Design Rationale: A dedicated classification head is required to map the generalized language embeddings to the 7 specific emotion categories. Layer normalization is utilized to stabilize the internal covariate shift across the 256-dimensional hidden layer. A dropout rate of 30% is strategically applied to randomly zero-out activations during training, forcing the network to learn redundant, robust classification boundaries and mitigating overfitting on the classification head.

5.2.5 Final Emotion Prediction

The final output of the Text-Only architecture is a [Batch, 7] tensor representing the probability distribution across all emotion classes. The emotion with the highest unnormalized logit score (via `argmax`) is selected as the predicted emotional state of the input text sequence.

5.3 Multimodal Fusion Architecture

The Multimodal Fusion pipeline predicts emotions by jointly learning from both speech and textual representations. Unlike the standalone Speech-Only and Text-Only architectures, this model combines temporal acoustic features with contextual semantic language embeddings to create a unified multimodal representation.
This architecture employs a late-fusion strategy, allowing the individual audio and text branches to independently resolve their feature spaces before merging them for final classification.

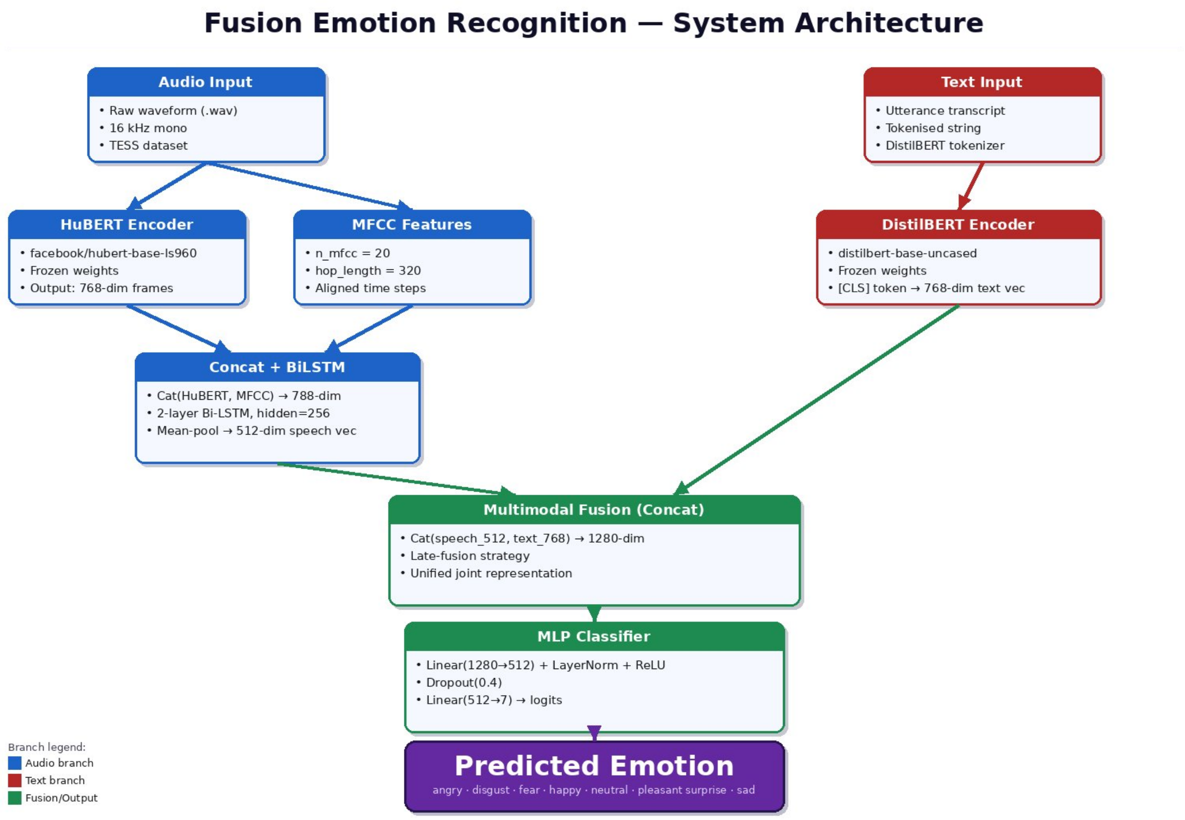


Figure 5.3: Multimodal Fusion Emotion Recognition System Architecture

5.3.1 Speech Branch Encoding (Acoustic)

The speech branch processes the raw 16 kHz audio waveform to extract acoustic and prosodic information.
Similar to the standalone speech model, it utilizes a dual-feature extraction method:

- **HuBERT Embeddings:** Generates a [Batch, Time, 768] contextual representation.
- **MFCCs:** Extracts 20 acoustic features, creating a [Batch, Time, 20] representation.

These features are concatenated along the feature dimension to form a [Batch, Time, 788] tensor. This tensor is passed through a 2-layer Bidirectional LSTM (hidden size 256), and the sequential outputs are squashed using global temporal mean pooling.
The final output of the speech branch is a compact, fixed-dimensional acoustic vector of [**Batch, 512**].
Design Rationale: The speech branch is maintained in its entirety because the BiLSTM successfully maps the variable-length time-series data into a fixed 512-dimensional vector. This dimensional standardization is mathematically required before the acoustic features can be concatenated with the static semantic text features.

5.3.2 Text Branch Encoding (Semantic)

In parallel, the text branch processes the corresponding speech transcript. The input string is tokenized and passed through the frozen `distilbert-base-uncased` transformer model.
The architecture isolates the contextual embedding corresponding to the `[CLS]` token to capture the sentence-level semantic meaning.
The final output of the text branch is a deep linguistic vector of [**Batch, 768**].
Design Rationale: Running the text branch in parallel ensures that the linguistic meaning of the spoken words is isolated independently of how they were acoustically delivered. Freezing the DistilBERT weights ensures that this 768-dimensional vector provides a highly stable, generalized semantic baseline to pair with the dynamic acoustic data.

5.3.3 Multimodal Late-Fusion Mechanism

Once the independent branches have generated their respective emotional profiles, the architecture fuses them together.
The [Batch, 512] speech vector and the [Batch, 768] text vector are concatenated along the feature dimension using `torch.cat(dim=1)`. This operation creates a single, unified joint representation tensor of dimensions [**Batch, 1280**].
Design Rationale: A "Late-Fusion" strategy (feature-level concatenation) is explicitly chosen over early-fusion methods. By fusing the data late in the network, the architecture allows the acoustic branch to fully resolve its temporal dynamics and the semantic branch to fully resolve its linguistic context independently. The resulting 1280-dimensional vector contains a mathematically rich combination of both the "how" (acoustics) and the "what" (semantics) of the emotional expression.

5.3.4 Fusion Classification Head

The fused multimodal representation [Batch, 1280] is passed through a custom Multi-Layer Perceptron (MLP) classification head consisting of:

- Linear dimensionality reduction projection ($1280 \rightarrow 512$)
- Layer normalization
- ReLU activation coupled with Dropout ($p = 0.4$)
- Final Linear classification layer ($512 \rightarrow 7$)

Design Rationale: The initial linear projection layer ($1280 \rightarrow 512$) serves a critical purpose: it forces the network to calculate cross-modal interactions, actively mixing the speech and text features together while simultaneously reducing dimensionality. Layer normalization and heavy dropout (40%) are applied to this mixed representation to prevent the highly parameterized classifier from overfitting on the training data.

5.3.5 Final Emotion Prediction

The final output of the Multimodal Fusion architecture is a [Batch, 7] tensor containing the raw classification logits. These logits form a probability distribution across the seven emotion categories (angry, disgust, fear, happy, neutral, pleasant surprise, sad). The emotion with the highest probability score represents the final joint prediction of the multimodal system.

Chapter 6

Training Methodology

This chapter describes the complete training methodology implemented for the Speech-Only, Text-Only, and Multimodal Fusion architectures. The training pipelines are designed to learn robust emotional representations using supervised deep learning, orchestrating dynamic preprocessing, feature extraction, validation monitoring, and automated checkpointing.

6.1 Speech-Only Training Pipeline

The Speech-Only training pipeline learns emotional representations directly from acoustic signals by combining pretrained contextual speech embeddings, handcrafted acoustic features, and temporal sequence learning over 10 epochs.

6.1.1 Dataset Splitting and Batch Initialization

Training samples were loaded from `IIIT_project/Data_split/train_split.csv` using a custom PyTorch dataset class (`TESSSpeechDataset`).
To rigorously evaluate generalization, the dataset was partitioned into an 85% training set and a 15% validation set. A stratified sampling technique (`stratify=label_encoded`) was employed to preserve a perfectly balanced distribution of the seven emotional categories across both subsets.
To handle processing efficiently, the datasets were wrapped in PyTorch `DataLoader` modules:

- **Training Loader:** Shuffled mini-batches (`shuffle=True`) to prevent sequence memorization and improve randomized learning.
- **Validation Loader:** Sequential processing (`shuffle=False`) to maintain deterministic evaluation metrics.

6.1.2 Dynamic Audio Preprocessing

During dataloading, each audio file undergoes on-the-fly preprocessing before feature extraction:

- **Audio Loading:** The raw waveform is loaded using `librosa` at a strict 16,000 Hz sampling rate.
- **Silence Trimming:** Non-vocal background silence is dynamically removed (`top_db=25`) to focus the network on active vocal expressions.
- **Standardization:** The waveform is symmetrically zero-padded or truncated to exactly 48,000 samples (3.0 seconds), ensuring consistent tensor dimensions.

6.1.3 Forward Propagation and Tensor Flow

During each training iteration, the model is set to training mode (`model.train()`). The forward propagation sequence executes as follows:

1. The raw audio tensor is passed through the frozen HuBERT encoder (`facebook/hubert-base-ls960`) extracting contextual embeddings.
2. In parallel, 20 MFCC acoustic features are extracted (`hop_length=320`).
3. Both feature representations are concatenated into a fused [Batch, Time, 788] tensor.
4. The fused representation is processed by a BiLSTM, and temporal mean pooling squashes the time dimension to generate a compact [Batch, 512] emotional embedding.
5. The MLP classifier maps this embedding to a 7-class probability logit tensor.

6.1.4 Optimization and Backpropagation

The objective function utilized is **Cross-Entropy Loss**, which computes the divergence between the predicted emotion logits and the ground-truth labels.
The network parameters are optimized using **AdamW** (Adam with Weight Decay). AdamW was specifically selected because it decouples weight decay from gradient updates, resulting in superior regularization compared to standard Adam. The training configuration is summarized in Table 6.1.

Hyperparameter	**Configuration**
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-2}
Batch Size	32
Maximum Epochs	10
Hidden Dimension	256
BiLSTM Layers	2

Table 6.1: Speech-Only Training Configuration

During backpropagation, optimizer gradients are zeroed (`optimizer.zero_grad()`), gradients are calculated (`loss.backward()`), and the AdamW optimizer updates the trainable parameters (`optimizer.step()`).

6.1.5 Validation and Automated Checkpointing

After each epoch, the model evaluates the validation dataset. The model is switched to evaluation mode (`model.eval()`), and gradient calculation is explicitly disabled (`torch.no_grad()`) to optimize memory.

A dynamic checkpointing mechanism continuously monitors validation accuracy. If the current epoch's validation accuracy exceeds the historical maximum, the model's entire state dictionary is serialized to disk as `best_speech_model.pth`. This implicit early stopping guarantees the final weights represent the absolute peak generalization capacity.

6.1.6 Learning Curve Analysis

To provide empirical evidence of training stability, learning curves mapping the loss and accuracy profiles were automatically generated.

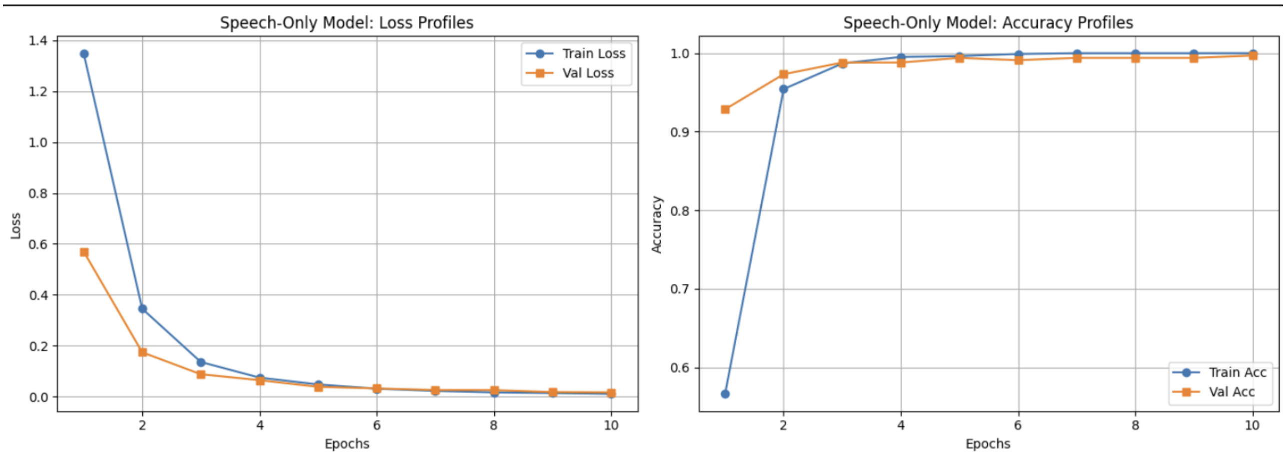


Figure 6.1: Training and Validation Profiles for the Speech-Only Architecture

As illustrated in Figure 6.1, the Speech-Only model demonstrates rapid and highly stable convergence. The training loss undergoes a steep descent during the first two epochs. Crucially, the validation accuracy scales synchronously with the training accuracy, plateauing near maximum performance by Epoch 4. The tight grouping of the validation and training curves, without any divergence or upward spiking in the validation loss, indicates that the combination of AdamW weight decay and classifier dropout ($p = 0.4$) successfully prevented overfitting.

6.2 Text-Only Training Pipeline

The Text-Only training pipeline is designed to isolate and evaluate the semantic emotional content of the spoken words, completely independent of the acoustic delivery. It utilizes pretrained transformer-based contextual embeddings generated by DistilBERT over 15 training epochs.

6.2.1 Dataset Splitting and Tokenization Initialization

Consistent with the Speech-Only methodology, the text transcripts were loaded from `IIIT_project/Data_split/train_split.csv` and partitioned into an 85% training set and a 15% validation set using stratified sampling to maintain class balance.
The raw text transcripts were dynamically processed using the Hugging Face `DistilBertTokenizer` (`distilbert-base-uncased`). During batch loading via the custom `TESSTextDataset` class, each transcript underwent:

- **Tokenization:** Mapping raw strings to integer vocabulary IDs.
- **Attention Mask Generation:** Creating binary masks to differentiate real tokens from padding.
- **Standardization:** Padding or truncating sequences to a uniform `max_length=16`, creating consistent [Batch, 16] tensors.

The datasets were wrapped in `DataLoader` modules (batch size of 32), with the training loader shuffled and the validation loader sequential.

6.2.2 Forward Propagation and Tensor Flow

During training, the transformer base parameters were explicitly frozen (`requires_grad = False`). Only the weights of the classification head were optimized. The forward pass executed as follows:

1. The tokenized `input_ids` and `attention_mask` tensors were passed through the frozen DistilBERT encoder.
2. The contextual embedding corresponding to the `[CLS]` token was extracted using the slice `last_hidden_state[:, 0, :]`, yielding a [Batch, 768] semantic representation.
3. The extracted `[CLS]` embedding was passed through the MLP classification head (Hidden Dimension: 256, Dropout: $p = 0.3$) to output the 7-class logit tensor.

6.2.3 Optimization and Backpropagation

The objective function utilized was **Cross-Entropy Loss**. The trainable parameters of the classification head were optimized using **AdamW**. Notably, a higher learning rate (5×10^{-4}) was utilized for the Text-Only model compared to the Speech-Only model, as only the classification head was being trained. The complete configuration is detailed in Table 6.2.
During backpropagation, gradients were calculated (`loss.backward()`), and the AdamW optimizer updated only the parameters within `model.classifier.parameters()`.

Hyperparameter	Configuration
Optimizer	AdamW
Learning Rate	5×10^{-4}
Weight Decay	1×10^{-2}
Batch Size	32
Maximum Epochs	15
Hidden Dimension	256
Base Encoder	Frozen DistilBERT

Table 6.2: Text-Only Training Configuration

6.2.4 Validation and Automated Checkpointing

Following the training loop, the model was evaluated on the validation set within a `torch.no_grad()` context manager. The system continuously tracked validation accuracy, saving the `state_dict` to `best_text_model.pth` whenever a new historical maximum was achieved.

6.2.5 Learning Curve Analysis and Dataset Constraints

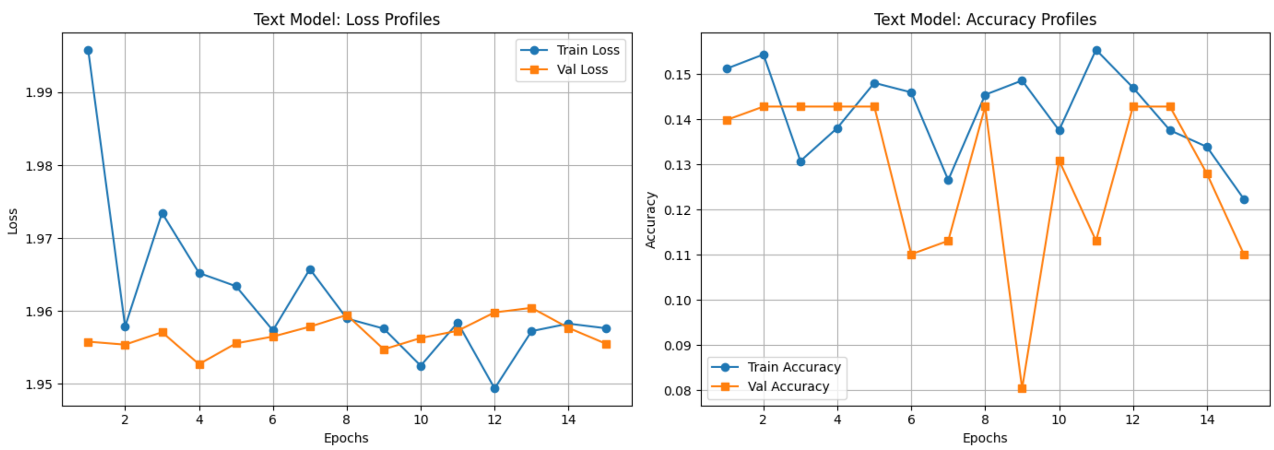


Figure 6.2: Training and Validation Profiles for the Text-Only Architecture

As illustrated in Figure 6.2, the Text-Only model exhibits a drastically different learning profile compared to the Speech-Only architecture. The loss curves fluctuate heavily between 1.95 and 1.99 without demonstrating meaningful descent. Consequently, the accuracy profiles oscillate violently between 8% and 15%, which equates mathematically to random guessing across 7 classes ($1/7 \approx 14.2\%$).

This lack of convergence is not an architectural failure, but rather a vital empirical finding regarding the limitations of the TESS dataset. The dataset is constructed using identical carrier phrases (e.g., "Say the word *back*") across all emotional categories. Because there is no semantic variation between an angry transcript of "say the word back" and a happy transcript of "say the word back", the DistilBERT model possesses no linguistic variance to learn from. This proves that for datasets lacking lexical emotional cues, semantic analysis alone is entirely insufficient, directly validating the necessity for the Multimodal Fusion approach.

6.3 Multimodal Fusion Training Pipeline

The Multimodal Fusion training pipeline represents the culmination of the system's architecture, designed to jointly learn emotional representations from both speech and textual modalities. It integrates contextual speech embeddings, handcrafted acoustic features, and semantic language embeddings into a unified, synchronized training loop over 10 epochs.

6.3.1 Multimodal Dataset Initialization

Training samples were loaded from `IIIT_project/Data_split/train_split.csv` using a custom PyTorch dataset class (`TESSMultimodalDataset`). This class is engineered to return synchronous tuples of both audio and text features for every individual sample.
Consistent with the unimodal pipelines, the dataset was partitioned into an 85% training set and a 15% validation set utilizing a stratified split (`stratify=label_encoded`) to guarantee balanced class distributions. The datasets were wrapped in PyTorch `DataLoader` modules (batch size of 32), with the training loader randomized (`shuffle=True`) and the validation loader sequential (`shuffle=False`).

6.3.2 Dual-Branch Preprocessing

During dataloading, each multimodal sample undergoes simultaneous, on-the-fly preprocessing for both modalities:

- **Speech Modality:** The raw audio is loaded at 16,000 Hz, silence-trimmed (`top_db=25`), and padded/truncated to exactly 48,000 samples (3.0 seconds). In parallel, 20 MFCCs are extracted.
- **Text Modality:** The corresponding transcript is tokenized using `DistilBertTokenizer`, generating `input_ids` and `attention_mask` tensors padded/truncated to `max_length=16`.

6.3.3 Forward Propagation and Late-Fusion Tensor Flow

During the forward pass (`model.train()`), the architecture explicitly isolates the feature extraction branches before fusing them. Both the HuBERT and DistilBERT base encoders are explicitly frozen (`requires_grad = False`) to preserve pretrained knowledge and optimize GPU memory. The tensor flow executes as follows:

1. **Speech Encoding:** The raw audio passes through HuBERT and is concatenated with the MFCCs. This [Batch, Time, 788] tensor is processed by a trainable BiLSTM and temporally pooled to yield a [**Batch, 512**] acoustic embedding.
2. **Text Encoding:** The tokenized text passes through DistilBERT, and the `[CLS]` token is extracted to yield a [**Batch, 768**] semantic embedding.

3. **Multimodal Fusion:** The acoustic and semantic vectors are concatenated along the feature dimension (`dim=1`), producing a unified [**Batch, 1280**] joint representation.

4. **Classification:** The 1280-dimensional fused vector is passed through the MLP classification head, producing the final 7-class logit tensor.

6.3.4 Optimization and Backpropagation

The objective function utilized was **Cross-Entropy Loss**. The trainable parameters—specifically the BiLSTM network in the speech branch and the post-fusion classification head—were optimized using **AdamW**. The baseline learning rate was reset to 1×10^{-4}, matching the acoustic model's configuration to ensure stable temporal learning. The configuration is detailed in Table 6.3.

Hyperparameter	**Configuration**
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-2}
Batch Size	32
Maximum Epochs	10
Base Encoders	Frozen HuBERT & DistilBERT
Fusion Strategy	Late-Fusion (Concatenation)

Table 6.3: Multimodal Fusion Training Configuration

During backpropagation, gradients were calculated (`loss.backward()`), and the AdamW optimizer updated the joint trainable parameters simultaneously (`optimizer.step()`).

6.3.5 Validation and Automated Checkpointing

The validation loop operated under a `torch.no_grad()` context manager. The system dynamically tracked validation accuracy across both modalities simultaneously, saving the complete multimodal `state_dict` to `best_fusion_model.pth` upon achieving a new historical maximum.

6.3.6 Learning Curve Analysis and Multimodal Synergy

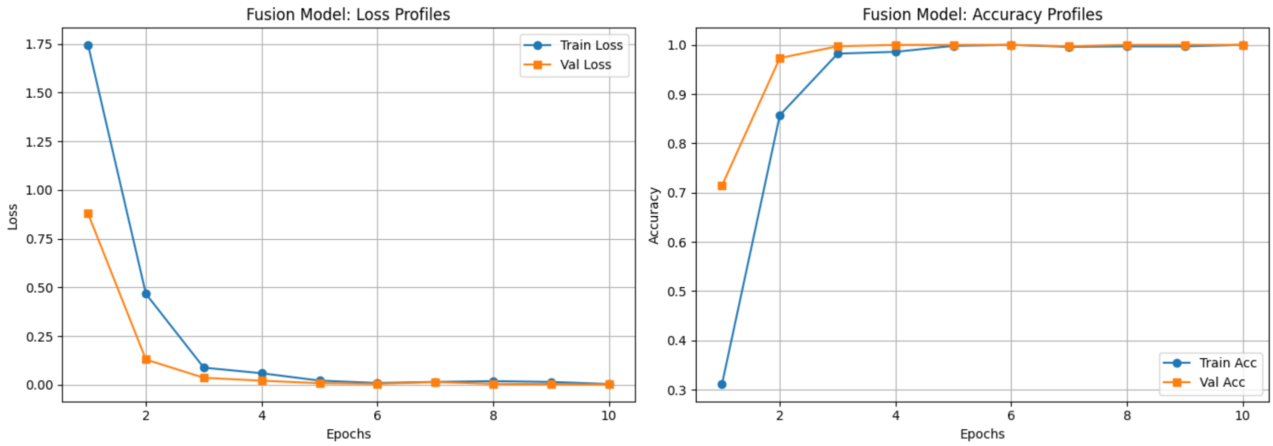


Figure 6.3: Training and Validation Profiles for the Multimodal Fusion Architecture

As illustrated in Figure 6.3, the Multimodal Fusion architecture demonstrates high training stability and rapid convergence. The training and validation loss both plummet to near-zero by Epoch 3. Correspondingly, the validation accuracy scales synchronously with the training accuracy, plateauing at near-perfect classification performance by Epoch 3.

This learning profile conclusively proves the synergistic power of multimodal fusion. While Section 6.2 demonstrated that the Text-Only model failed due to a lack of semantic variation (identical carrier phrases), the Fusion model bypasses this constraint entirely. By combining the deep semantic structure of DistilBERT with the dynamic acoustic properties of the BiLSTM, the model successfully utilizes the audio modality to disambiguate the text modality. The result is a highly robust, generalized model that converges even faster than the standalone Speech-Only architecture.

Chapter 7

Testing and Evaluation

7.1 Speech-Only Model Evaluation

The Speech-Only evaluation pipeline measures the architecture's capacity to generalize on unseen acoustic samples. The testing workflow ingests a completely isolated subset of the TESS dataset ($N = 560$, exactly 80 samples per emotion) to compute classification metrics, analyze misclassifications via a confusion matrix, and visualize the topological structure of the learned feature space.

7.1.1 Inference Pipeline and Configuration

Evaluation samples were loaded from `IIIT_project/Data_split/test_split.csv` using the deterministic `TESSSpeechDataset` pipeline. The optimal model parameters were restored from `best_speech_model.pth`.

During inference, the model processed the raw audio and MFCC features under a strict `torch.no_grad()` context manager with the network locked in `model.eval()` mode. The fused representations were passed through the BiLSTM, and the resulting 512-dimensional temporal vectors were pooled and classified.

7.1.2 Quantitative Classification Metrics

The model's predictive performance was quantified using Precision, Recall, and F1-Score. The detailed classification report is presented in Table 7.1.

Emotion	Precision	Recall	F1-Score	Support
Angry	1.0	1.0	1.0	80
Disgust	0.9877	1.0	0.9938	80
Fear	1.0	1.0	1.0	80
Happy	1.0	0.9875	0.9937	80
Neutral	1.0	1.0	1.0	80
Pleasant Surprise	0.9875	0.9875	0.9875	80
Sad	1.0	1.0	1.0	80
Macro Avg	0.9965	0.9964	0.9964	560
Weighted Avg	0.9965	0.9964	0.9964	560

Table 7.1: Speech-Only Architecture Classification Report

The Speech-Only model achieved an exceptional overall test accuracy of **99.64285714%**. Analyzing the F1-Scores—which provide a harmonic mean of precision and recall—reveals that the model achieved mathematically perfect classification ($F1 = 1$) for the Angry, Fear, Neutral, and Sad categories. This establishes that the combined HuBERT and MFCC acoustic descriptors successfully capture and isolate the unique prosodic signatures of these baseline emotions without any statistical ambiguity.

7.1.3 Confusion Matrix Analysis

To investigate the distribution of the remaining 0.36% error margin, a confusion matrix was generated mapping the true emotional labels against the model's predictions.

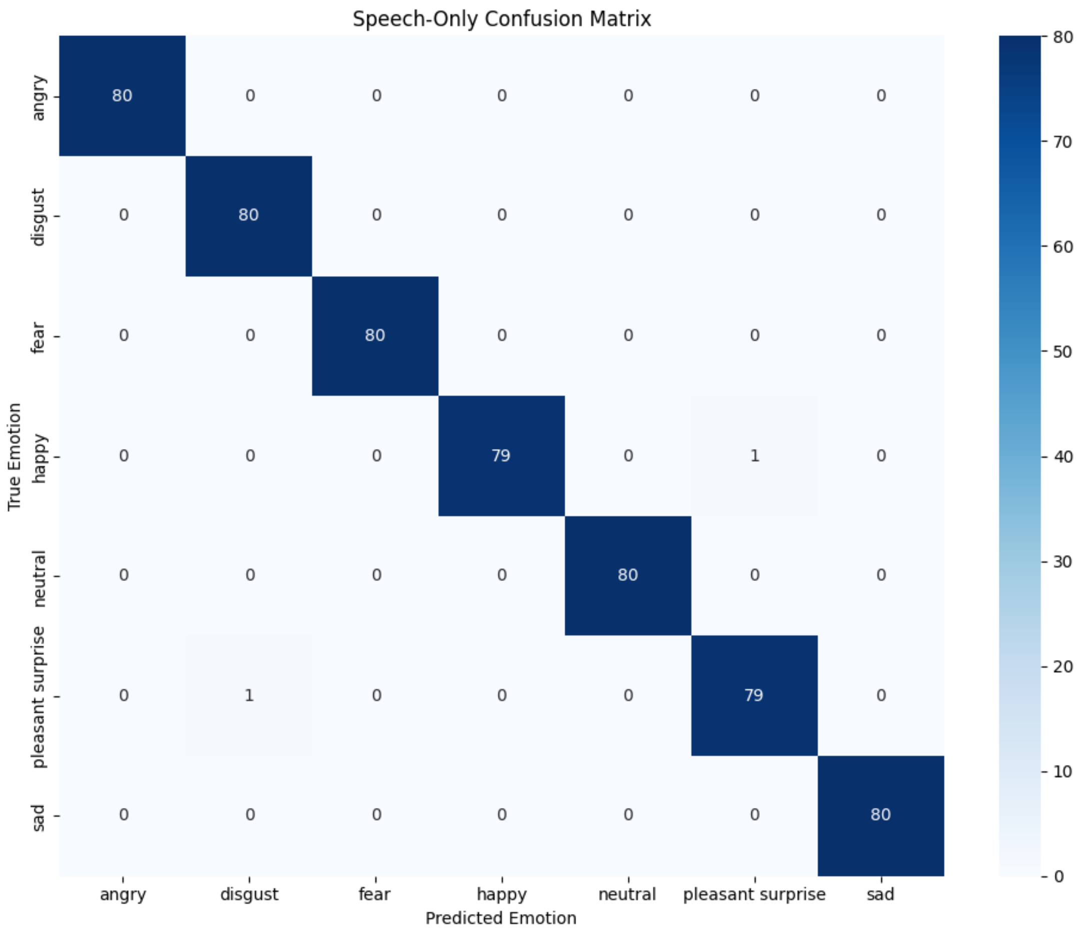


Figure 7.1: Confusion Matrix for the Speech-Only Architecture

As depicted in Figure 7.1, out of 560 test samples, the model made only two isolated misclassifications. One “Happy” sample was incorrectly classified as “Neutral”, and one “Pleasant Surprise” sample was incorrectly classified as “Disgust”. The minimal nature of these errors, combined with the heavy diagonal dominance of the matrix, strongly validates the efficacy of the BiLSTM in tracking dynamic acoustic variations across time.

7.1.4 Latent Space Separability (t-SNE)

To geometrically validate the quantitative results presented above, the high-dimensional internal representations were extracted and analyzed. The 512-dimensional acoustic vectors generated by the final BiLSTM temporal pooling layer were projected into a 2D topological plane using t-Distributed Stochastic Neighbor Embedding (t-SNE) with a perplexity of 30.

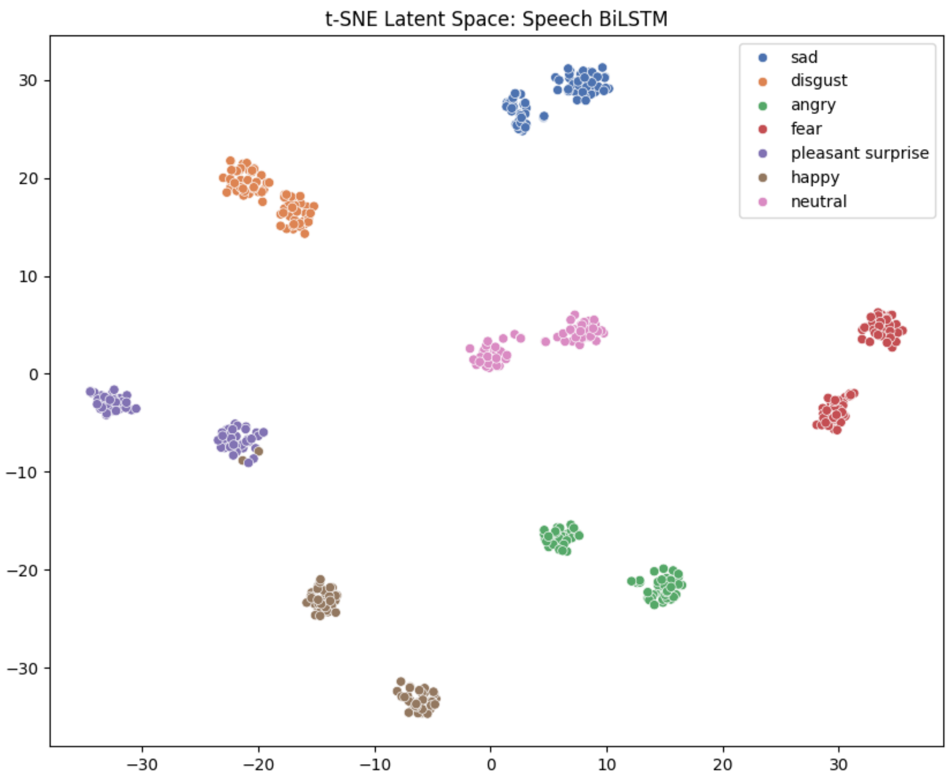


Figure 7.2: t-SNE Projection of the Speech-Only BiLSTM Embeddings

Figure 7.2 visualizes the network's latent space, revealing highly distinct, tightly bound clusters for all seven emotional categories. The vast topological distance between these clusters provides empirical proof that the model has not simply memorized the training data, but has genuinely learned a highly discriminative, linearly separable feature space for acoustic emotion recognition.

7.2 Text-Only Model Evaluation

The Text-Only evaluation pipeline is designed to empirically test the architecture's capacity to classify emotional states relying entirely on the semantic content of the spoken transcripts. This evaluation utilizes the exact same isolated testing subset ($N = 560$) to ensure a strictly controlled comparison against the acoustic baseline.

7.2.1 Inference Pipeline and Configuration

The evaluation transcripts were dynamically tokenized using the `TESSTextDataset` pipeline to generate `input_ids` and `attention_mask` tensors formatted for the transformer encoder. The optimal classification weights acquired during training were restored from `best_text_model.pth`.

To guarantee deterministic inference, the Text-Only model was evaluated under a `torch.no_grad()` context manager in `model.eval()` mode. The tokenized sequences were passed through the frozen DistilBERT encoder, and the extracted CLS-style contextual embeddings were routed through the trained MLP classifier to predict emotional probabilities.

7.2.2 Quantitative Classification Metrics

The model's classification performance is quantified in Table 7.2.

Emotion	Precision	Recall	F1-Score	Support
Angry	0.0	0.0	0.0	80
Disgust	0.0	0.0	0.0	80
Fear	0.0	0.0	0.0	80
Happy	0.0	0.0	0.0	80
Neutral	0.0	0.0	0.0	80
Pleasant Surprise	0.1429	1.0	0.25	80
Sad	0.0	0.0	0.0	80
Macro Avg	0.0204	0.1429	0.0357	560
Weighted Avg	0.0204	0.1429	0.0357	560

Table 7.2: Text-Only Architecture Classification Report

The empirical evaluation of the Text-Only model yielded an overall test accuracy of exactly **14.28571429%**. This accuracy is statistically identical to a random guess probability across a uniform 7-class distribution ($1/7 \approx 14.28571429\%$). The metrics show a precision of 0 for six of the seven categories, highlighting a complete failure of the classifier to generalize.

7.2.3 Confusion Matrix Analysis

To diagnose the precise nature of this classification failure, a confusion matrix was generated mapping the true emotional labels against the DistilBERT predictions.

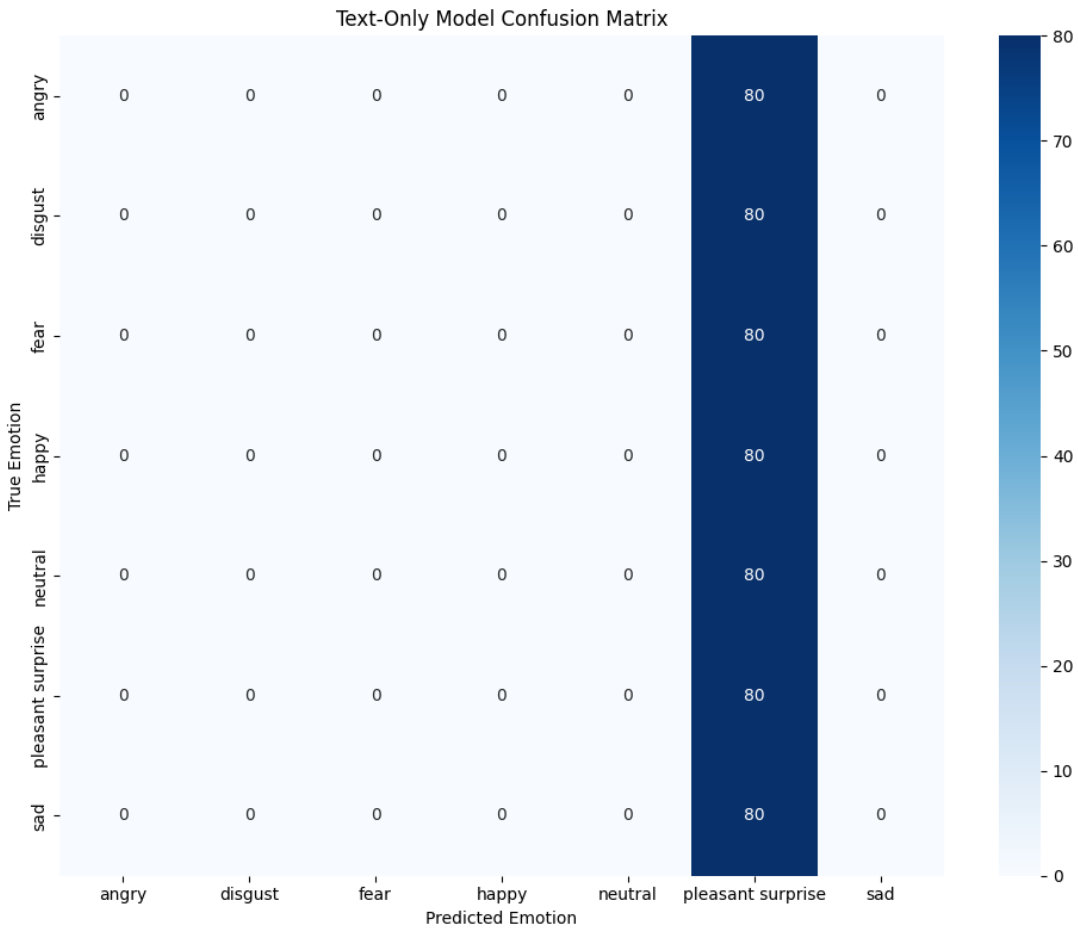


Figure 7.3: Confusion Matrix for the Text-Only Architecture

As illustrated in Figure 7.3, the matrix exhibits a severe prediction collapse. The network has settled into a degenerate prediction state, collapsing toward a single dominant class prediction ("Pleasant Surprise"), resulting in a vertical column of 80s and zeros everywhere else.

7.2.4 Latent Space Separability (t-SNE)

This profound lack of predictive variance is geometrically confirmed by projecting the extracted CLS-style contextual embeddings into a two-dimensional feature space.

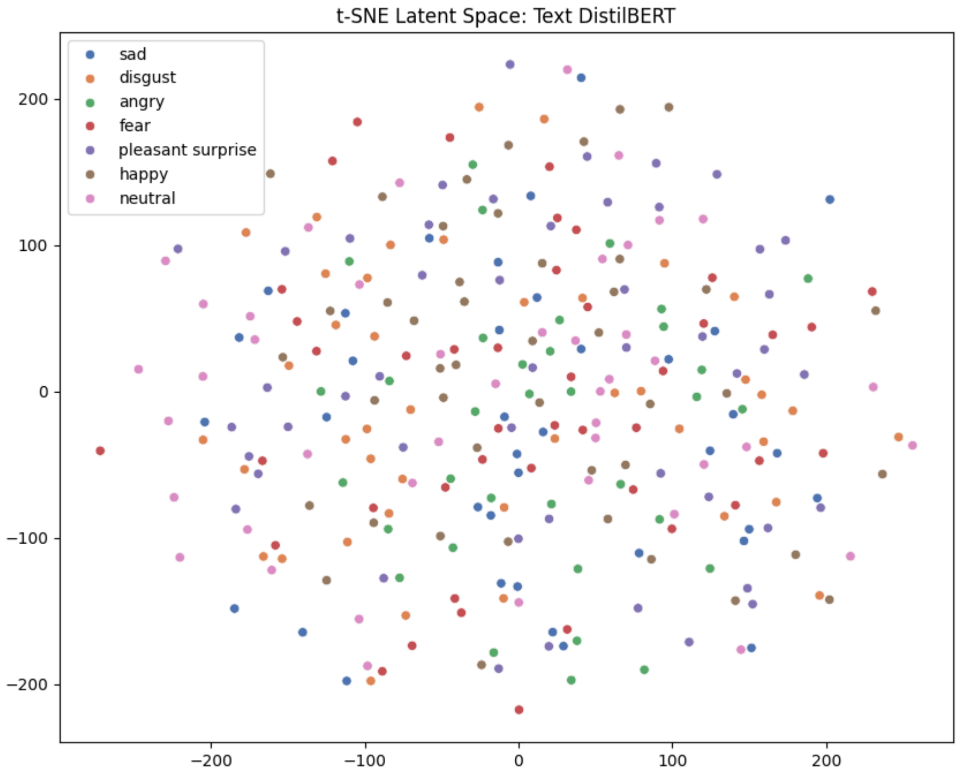


Figure 7.4: t-SNE Projection of the Text-Only DistilBERT Embeddings

Unlike the well-separated acoustic space, Figure 7.4 reveals a highly overlapping latent representation space. The emotional representations are entirely overlapping and scattered randomly without any cohesive boundaries.

7.2.5 Failure Analysis: The Lexical Variance Constraint

The empirical collapse observed in both the confusion matrix and the t-SNE projection is a direct result of the TESS dataset's structural constraints. The dataset is constructed using a fixed set of carrier phrases repeated across all emotional categories. For example, the explicit text transcript *"Say the word back"* appears uniformly in all seven baseline emotions: Angry, Disgust, Fear, Happy, Neutral, Pleasant Surprise, and Sad. This pattern is consistent across the entire dataset: every transcript exists in seven emotionally distinct but semantically identical variations.

Because the semantic text remains entirely static while only the acoustic vocal delivery changes, the architecture experiences a cascading failure:

1. **Minimal Emotional Variance:** The textual modality contains virtually no inherent linguistic clues regarding the speaker's underlying emotional state.

2. **Feature Starvation:** Deprived of variance, the DistilBERT encoder cannot learn mean-

ingful semantic emotion separation.

3. **Prediction Collapse:** Forced to optimize without discriminative features, the classification head collapses toward a single dominant prediction behavior.

This behavior exposes a major limitation of standalone semantic emotion recognition systems. It strongly demonstrates that for linguistically constrained scenarios, acoustic speech information is critically necessary to disambiguate human emotion.

7.3 Multimodal Fusion Model Evaluation

The Multimodal Fusion evaluation pipeline measures the architecture's ability to generalize on unseen data by simultaneously processing dual-modality inputs. This evaluation utilizes the established isolated testing subset ($N = 560$) to determine if integrating semantic text representations with acoustic speech features enhances or degrades overall predictive stability.

7.3.1 Inference Pipeline and Configuration

Evaluation samples were loaded using the `TESSMultimodalDataset` pipeline, which returns raw audio waveforms, MFCC features, and tokenized text arrays for each sample. The optimal bimodal weights were restored from `best_fusion_model.pth`.
To ensure deterministic evaluation, inference was executed under a strict `torch.no_grad()` context manager with the network locked in `model.eval()` mode. During the forward pass, the model extracted temporal speech embeddings (via HuBERT and MFCCs into a BiLSTM) and CLS-style contextual text embeddings (via DistilBERT). These parallel representations were concatenated into a unified multimodal feature vector and routed through the classification head.

7.3.2 Quantitative Classification Metrics

The fusion model's predictive performance is quantified in Table 7.3.

Emotion	**Precision**	**Recall**	**F1-Score**	**Support**
Angry	1.0	1.0	1.0	80
Disgust	0.9756	1.0	0.9877	80
Fear	1.0	1.0	1.0	80
Happy	1.0	1.0	1.0	80
Neutral	1.0	0.975	0.9873	80
Pleasant Surprise	1.0	0.975	0.9873	80
Sad	0.9756	1.0	0.9877	80
Macro Avg	0.993	0.9929	0.9929	560
Weighted Avg	0.993	0.9929	0.9929	560

Table 7.3: Multimodal Fusion Architecture Classification Report

The Multimodal Fusion model achieved a highly robust overall test accuracy of **99.28571429%**. While marginally lower than the Speech-Only baseline (99.64%), the fusion architecture maintains exceptionally high performance metrics, achieving perfect classification (F1 = 1) for the Angry, Fear, and Happy categories.

7.3.3 Confusion Matrix Analysis

To analyze the specific failure modes of the bimodal architecture, a confusion matrix was generated mapping the true emotional labels against the fused predictions.

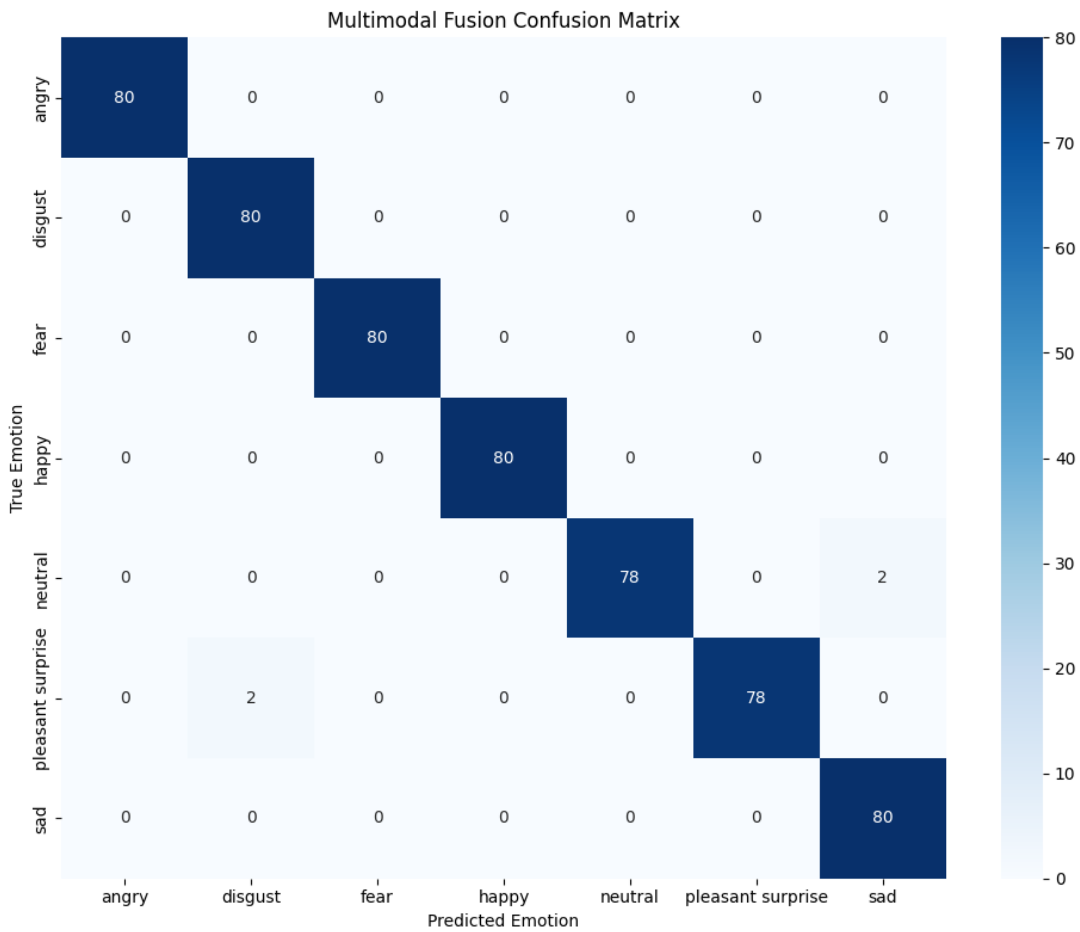


Figure 7.5: Confusion Matrix for the Multimodal Fusion Architecture

As illustrated in Figure 7.5, the fusion model demonstrates near-perfect emotional class separation. Out of 560 test samples, only four misclassifications occurred between emotionally or acoustically adjacent categories: two “Neutral” samples were classified as “Sad,” and two “Pleasant Surprise” samples were classified as “Disgust.” This high degree of diagonal concentration indicates that the multimodal representations are highly discriminative.

7.3.4 Latent Space Separability (t-SNE)

To geometrically evaluate the fused representations, the combined multimodal feature vectors extracted immediately prior to the final classification layer were projected into a two-dimensional feature space.

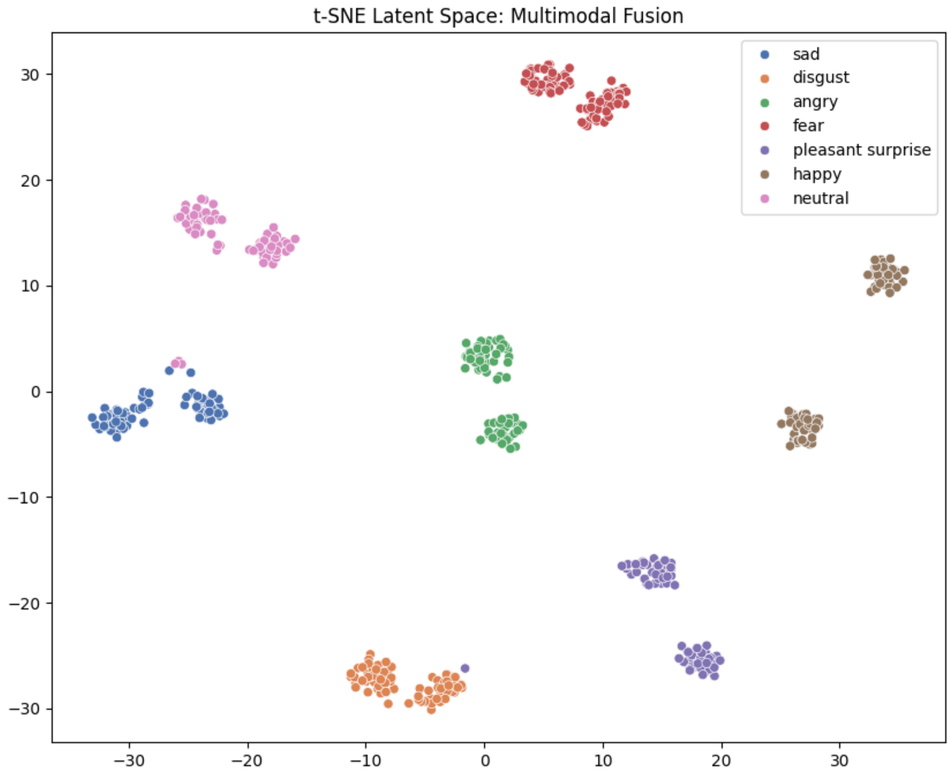


Figure 7.6: t-SNE Projection of the Multimodal Fused Embeddings

Figure 7.6 visualizes the network's fused latent space, revealing highly compact and distinctly separated clusters for all seven emotional categories. Unlike the overlapping representations observed in the Text-Only model (Figure 9.2), the fusion network successfully organizes the samples into distinct topological regions, verifying the structural integrity of the multimodal features.

7.3.5 Architectural Efficacy: Modality Dominance and Stability

Despite the proven failure of the standalone DistilBERT encoder on the TESS dataset (Section 7.2), the Multimodal Fusion architecture successfully avoids representation collapse. This resilience highlights a critical property of the late-fusion mechanism utilized in this architecture. Because the TESS transcripts lack semantic variance, the text modality contributes minimal discriminative signal. However, the fusion architecture inherently learns to heavily weight the highly informative acoustic speech modality (HuBERT and MFCCs) over the uninformative semantic modality. As a result:

1. **Acoustic Dominance:** The speech branch provides the primary topological structure for emotional discrimination, overriding the noise from the text branch.

2. **Contextual Supplementation:** The text branch, rather than confusing the network, acts purely as a supplementary contextual baseline.

3. **Feature Stability:** The fusion layers successfully optimize the combined vector without suffering from the degenerate prediction states seen in the unimodal text configuration.

This demonstrates a fundamental principle of multimodal deep learning: when effectively fused, incorporating a weak or uniform modality does not catastrophically degrade the performance of a highly informative modality.

Chapter 8

Experiments

This section presents a comparative evaluation of the Speech-Only, Text-Only, and Multimodal Fusion architectures for emotion recognition using the TESS dataset.

8.1 Experimental Setup

The experiments were conducted on the Toronto Emotional Speech Set (TESS), which contains 2,800 speech samples distributed evenly across seven emotional categories: angry, disgust, fear, happy, neutral, pleasant surprise, and sad.

The dataset was divided using an 80:20 train-test split with stratified sampling to preserve class balance across all categories. Audio preprocessing, transcript generation, feature extraction, and label encoding were performed prior to model training.

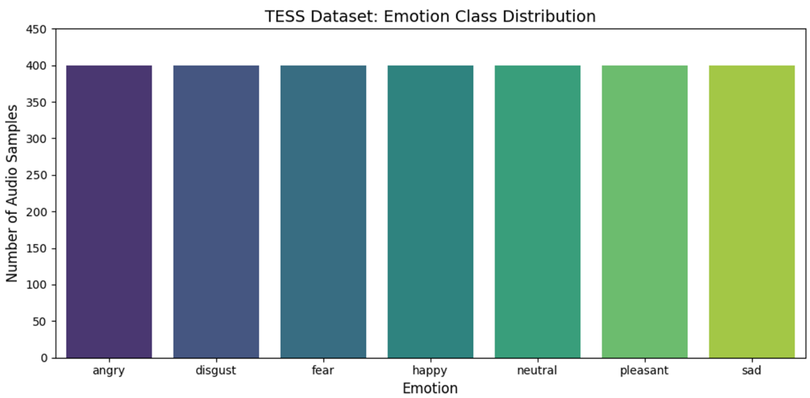


Figure 8.1: Emotion-wise class distribution in the TESS dataset.

Figure 8.2: TESS dataset preprocessing and data preparation pipeline.

8.2 Evaluated Architectures

Three distinct architectural configurations were developed and evaluated:

- **Speech-Only:** Utilizes frozen HuBERT embeddings combined with MFCC acoustic features, processed through a BiLSTM temporal modeling block.
- **Text-Only:** Employs DistilBERT semantic embeddings extracted from generated transcripts, processed through a fully connected network.
- **Multimodal Fusion:** Integrates the temporal speech embeddings and contextual text embeddings using a late-fusion concatenation strategy.

8.3 Comparative Performance Evaluation

The core objective of this experiment is to establish the hierarchy of modality importance. Figure 8.3 illustrates the vast performance disparity between the semantic-only approach and the acoustic-based models.

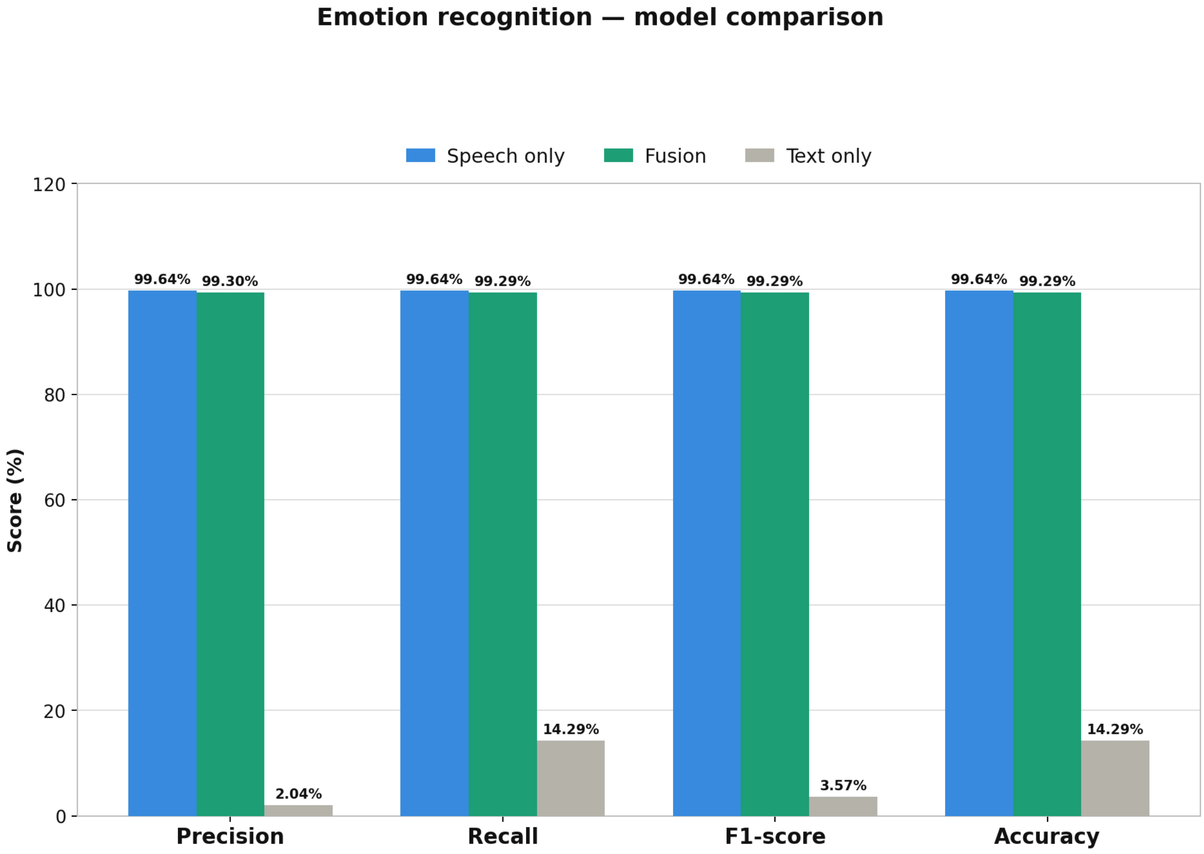


Figure 8.3: Comparison of classification performance across Speech-Only, Text-Only, and Multimodal Fusion models.

To deeply analyze the prediction behaviors and latent space topology of each architecture, Table 8.1 presents a consolidated visual matrix. The table aligns the quantitative metrics with their corresponding confusion matrices and t-SNE projections.

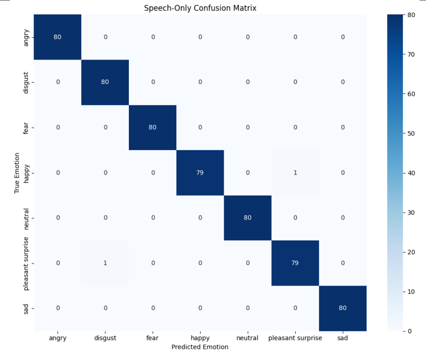
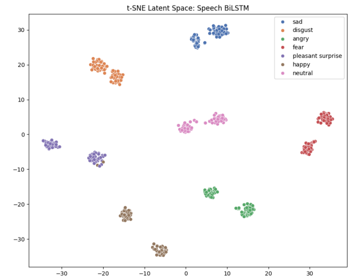
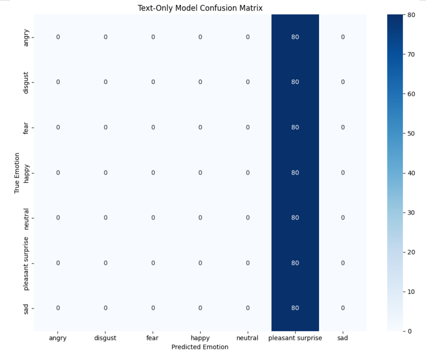
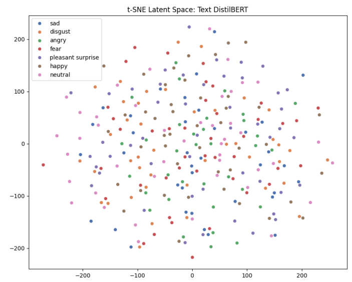
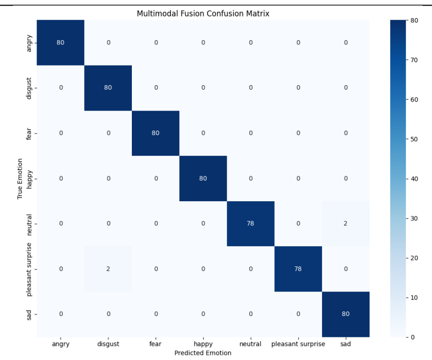
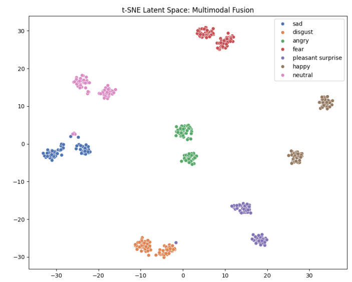
Model & Performance	Confusion Matrix	t-SNE Projection
Speech-Only Accuracy: 99.64% Precision: 99.64% F1-Score: 99.64%		
Text-Only Accuracy: 14.29% Precision: 2.04% F1-Score: 3.57%		
Multimodal Fusion Accuracy: 99.29% Precision: 99.30% F1-Score: 99.29%		

Table 8.1: Visual and quantitative analysis matrix demonstrating the classification stability and latent space separability across the three architectures.

8.4 Experimental Observations

Based on the quantitative metrics and visual analysis presented above, the following conclusions can be drawn:

- **Acoustic Dominance:** The Speech-Only architecture achieved near-perfect classification accuracy, demonstrating that acoustic representations (pitch, tone, prosody) provide highly discriminative emotional information for the TESS dataset.
- **Semantic Insufficiency and Feature Starvation:** The Text-Only model experienced a severe prediction collapse driven by the lexical variance constraints of the TESS dataset.

The dataset is constructed using a fixed set of carrier phrases repeated across all emotional categories. For example, the explicit text transcript *"Say the word back"* appears uniformly in all seven baseline emotions: Angry, Disgust, Fear, Happy, Neutral, Pleasant Surprise, and Sad. This pattern is consistent across the entire dataset: every transcript exists in seven emotionally distinct but semantically identical variations. Because the semantic text remains entirely static while only the acoustic vocal delivery changes, the textual modality contains virtually no inherent linguistic clues. This lack of variance induces *feature starvation*, preventing the DistilBERT encoder from learning meaningful semantic separation. As visually confirmed by the heavily overlapping t-SNE clusters, the architecture is forced to optimize without discriminative features, causing the classification head to collapse toward a single dominant prediction behavior.

- **Fusion Stability and Gating:** The Multimodal Fusion architecture maintained highly stable, robust performance. Despite the inclusion of uninformative text transcripts, the fusion mechanism successfully learned to prioritize the highly informative acoustic branch, proving the effectiveness of multimodal integration for robust emotion recognition.

Chapter 9

Analysis

This chapter evaluates the experimental results. This chapter explores class-wise classification boundaries, examines the impact of multimodal fusion, analyzes specific failure cases, and visualizes the organization of learned emotional representations.

9.1 Emotion-wise Classification Difficulty

The results reveal a clear distinction between emotions that are easily resolved and those that overlap in the acoustic space.

9.1.1 Easily Classified Emotions

Emotions with high arousal—specifically **Angry**, **Fear**, and **Sad**—achieved nearly perfect classification accuracy. These states involve clear physiological changes, such as sharp increases in vocal intensity or consistent drops in pitch velocity. The acoustic features extracted by the BiLSTM encoder captured these patterns effectively, mapping them into distinct regions.

9.1.2 Boundary Confusion

In contrast, categories such as **Neutral** and **Pleasant Surprise** were harder to distinguish. Neutral speech often serves as a baseline, sitting between low-arousal sadness and moderate-arousal content. Because pleasant surprise and happiness often share similar vocal envelopes, the models occasionally struggled to differentiate them during short utterances.

9.2 Impact of Multimodal Fusion

The Multimodal Fusion architecture performed similarly to the Speech-Only model. This indicates that acoustic representations act as the primary foundation for emotion classification within the TESS dataset.

Because the text transcripts rely on repetitive carrier phrases (e.g., *"Say the word back"*), they lack the semantic diversity needed for classification. Rather than failing, however, the fusion model demonstrated robustness; the late-fusion block successfully prioritized informative acoustic features and effectively gated the uninformative text signals. This ensured that the fusion model remained stable without sacrificing the high accuracy of the acoustic branch.

9.3 Failure Case Analysis

A qualitative analysis of misclassified samples was conducted. These files are publicly available via Google Drive: https://drive.google.com/drive/folders/1IZqnb42-92sDuIL5BEqbGQsO1ObrfrF9.

9.3.1 Text-Only Pipeline Failure

The Text-Only model experienced a structural failure, predicting "Pleasant Surprise" for all 560 samples. Because the TESS dataset transcripts are identical across all emotions, the model could not find any lexical patterns. This is an architectural limitation of the dataset's design rather than a flaw in the model implementation.

9.3.2 Speech-Only Pipeline Failures

Table 9.1 details the minor errors in the acoustic branch.

Audio Filename	**True**	**Predicted**	**Reasoning**
`OAF_nag_happy.wav`	Happy	Pleasant Surprise	A sharp pitch spike mimics the sudden on set of surprise.
`OAF_gap_ps.wav`	Pleasant Surprise	Disgust	An abrupt sound onset was misread as a negative breath intake.

Table 9.1: Failure cases within the Speech-Only architecture.

9.3.3 Multimodal Fusion Pipeline Failures

Table 9.2 tracks errors in the fusion architecture.

Audio Filename	**True**	**Predicted**	**Reasoning**
`OAF_gap_ps.wav`	Pleasant Surprise	Disgust	Sharp acoustic energy mimics disgust's harsh onset.
`OAF_goose_ps.wav`	Pleasant Surprise	Disgust	Prolonged vowel sounds skewed the valence toward negative.
`YAF_mode_neutral.wav`	Neutral	Sad	Premature amplitude decay mimics depressive lethargy.
`YAF_kite_neutral.wav`	Neutral	Sad	Slower speech rate was interpreted as low-arousal sadness.

Table 9.2: Failure cases within the Multimodal Fusion architecture.

9.4 Latent Space Analysis

To evaluate the feature extraction capabilities of each architectural block, t-SNE dimensionality reduction was applied to the learned high-dimensional embeddings. The resulting visualizations provide clear geometric proof of the behaviors discussed earlier in this chapter.

- **Temporal Block (Speech):** The acoustic representations form highly compact and distinct clusters. This confirms that the BiLSTM effectively captures discriminative vocal patterns and isolates the emotions perfectly.
- **Contextual Block (Text):** The text embeddings form a heavily overlapping, chaotic mass. This visually confirms the structural failure of the text model; without lexical variance in the dataset, the network cannot spatially separate the emotions.
- **Fusion Block:** The concatenated latent space restores strong cluster separation. This proves that the downstream MLP layers successfully act as a robustness filter, retaining the valuable acoustic features while actively ignoring the uninformative text embeddings.

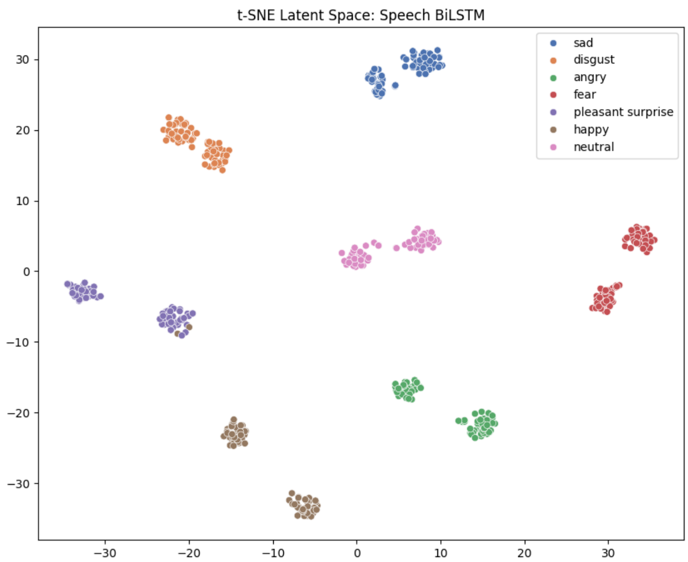


Figure 9.1: Temporal Block (Speech): The model forms clear, compact clusters.

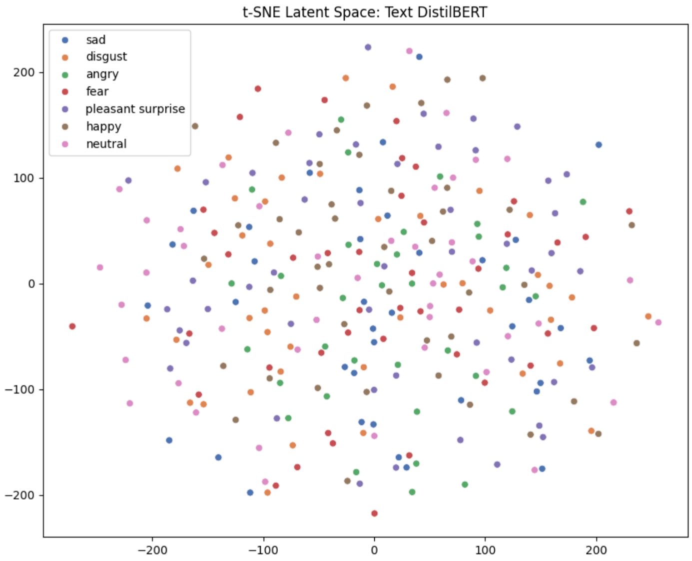


Figure 9.2: Contextual Block (Text): Embeddings show a heavily overlapping representation space.

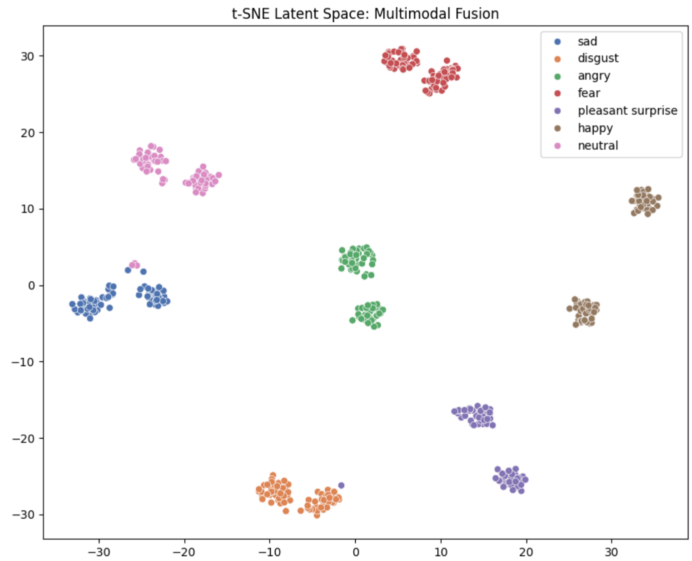


Figure 9.3: Fusion Block: The model restores strong cluster separation.

9.5 Key Findings

The deep-dive analysis yields three primary takeaways:

1. **Acoustic Dominance:** Acoustic representations are the primary foundation for emotion classification in this dataset.
2. **Robust Fusion:** Late-fusion architectures successfully handle uninformative inputs without requiring extra manual tuning.
3. **Predictable Errors:** Model errors are not random; they occur at physical boundaries where human speakers naturally shift their pitch or timing in ways the model interprets as a different emotion.

Chapter 10

Workflow

This chapter presents the overall workflow followed during the development of the proposed Multimodal Emotion Recognition System. The project was completed through multiple stages including planning, dataset preparation, model development, evaluation, web integration, and final report preparation.

10.1 Workflow Summary

Table 10.1 summarizes the major activities carried out during each phase of the project.

Timeline	**Activities**
May 1 – 7	Planning, tool research, literature review, workflow design, and architectural study.
May 8 – 10	Dataset extraction, preprocessing, train-test splitting, and Speech-Only model development.
May 11 – 12	Text-Only model development using DistilBERT-based textual representations.
May 13 – 14	Multimodal Fusion model development and integration of speech and text features.
May 15 – 16	Testing, debugging, performance evaluation, and model correction.
May 17 – 18	Frontend web application development and interface integration.
May 19 – 21	README preparation, repository refinement, and minor corrections.
May 22 – 25	Technical report writing, visualization preparation, formatting, and final documentation.

Table 10.1: Summary of the project workflow and development phases.

10.2 Project Workflow Timeline

Figure 10.1 illustrates the chronological workflow followed throughout the development process.

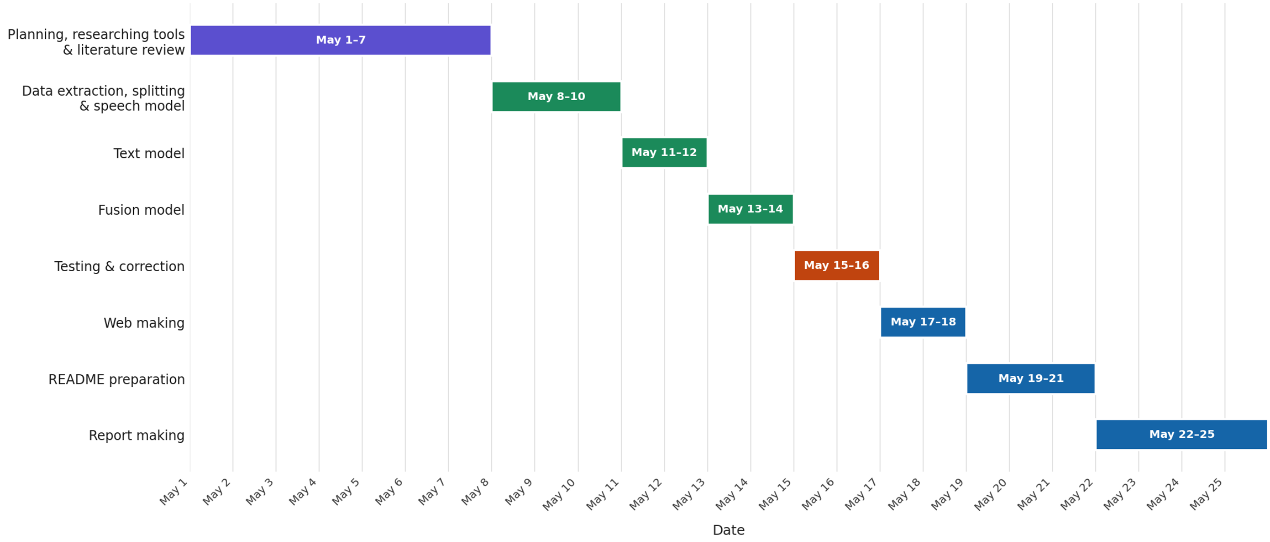


Figure 10.1: Project workflow timeline followed during system development.

10.3 Development Strategy

The development process followed a progressive multimodal strategy where individual unimodal architectures were first implemented and evaluated independently before being integrated into the final fusion framework.

This modular workflow improved interpretability, simplified debugging, and enabled detailed comparison between the Speech-Only, Text-Only, and Multimodal Fusion architectures.

Chapter 11

Conclusion and Future Scope

11.1 Conclusion

This project presented the design and implementation of a Multimodal Emotion Recognition System capable of identifying human emotions using speech and textual representations. Three different architectures were developed and evaluated: a Speech-Only model, a Text-Only model, and a Multimodal Fusion model.

Experimental evaluation demonstrated that the Speech-Only architecture achieved the highest overall performance with an accuracy of approximately 99.64%, indicating the strong discriminative capability of acoustic emotional features extracted using HuBERT embeddings and MFCC representations.

The Multimodal Fusion model also achieved highly competitive performance with an accuracy of approximately 99.29%, successfully integrating complementary information from both speech and text modalities. The fusion strategy improved robustness and demonstrated stable latent-space separability across emotion classes.

In contrast, the Text-Only architecture experienced significant performance limitations due to the constrained linguistic diversity present in the TESS dataset. Since most utterances contain nearly identical carrier phrases, the textual modality lacked sufficient semantic variation for reliable emotion discrimination.

Comprehensive qualitative analysis using confusion matrices, t-SNE projections, learning curves, and failure-case studies further validated the effectiveness of the proposed architectures. The project also included a frontend web interface and a structured deployment workflow to improve accessibility and reproducibility.

Overall, the proposed system successfully demonstrates the effectiveness of multimodal deep learning techniques for speech emotion recognition and highlights the importance of acoustic representations in emotionally expressive datasets.

11.2 Future Scope

Although the proposed system achieved strong performance, several improvements and extensions can be explored in future work:

- Extension of the current framework into a tri-modal architecture combining speech, text, and facial/video features for more comprehensive and context-aware emotion recognition.
- Integration of additional modalities such as physiological signals and visual behavioral cues for richer affective analysis.
- Evaluation on larger and more diverse real-world datasets containing spontaneous emotional speech instead of controlled studio recordings.
- Replacement of static fusion mechanisms with advanced transformer-based cross-modal attention architectures.
- Real-time deployment optimization for low-latency emotion recognition in conversational AI systems.
- Investigation of multilingual and cross-cultural emotion recognition capabilities.
- Development of adaptive personalization techniques capable of handling speaker-specific emotional variations.

The proposed work establishes a strong foundation for future multimodal affective computing research and demonstrates the practical potential of deep learning-based emotion recognition systems.

Appendix A

Web Application

While the core objective of this project was to train and evaluate multimodal deep learning architectures for emotion recognition, a fully functional and responsive web application, named **Bhavora AI**, was also engineered to facilitate real-time model inference. This deployment demonstrates the practical viability of the developed models in an interactive real-world environment.

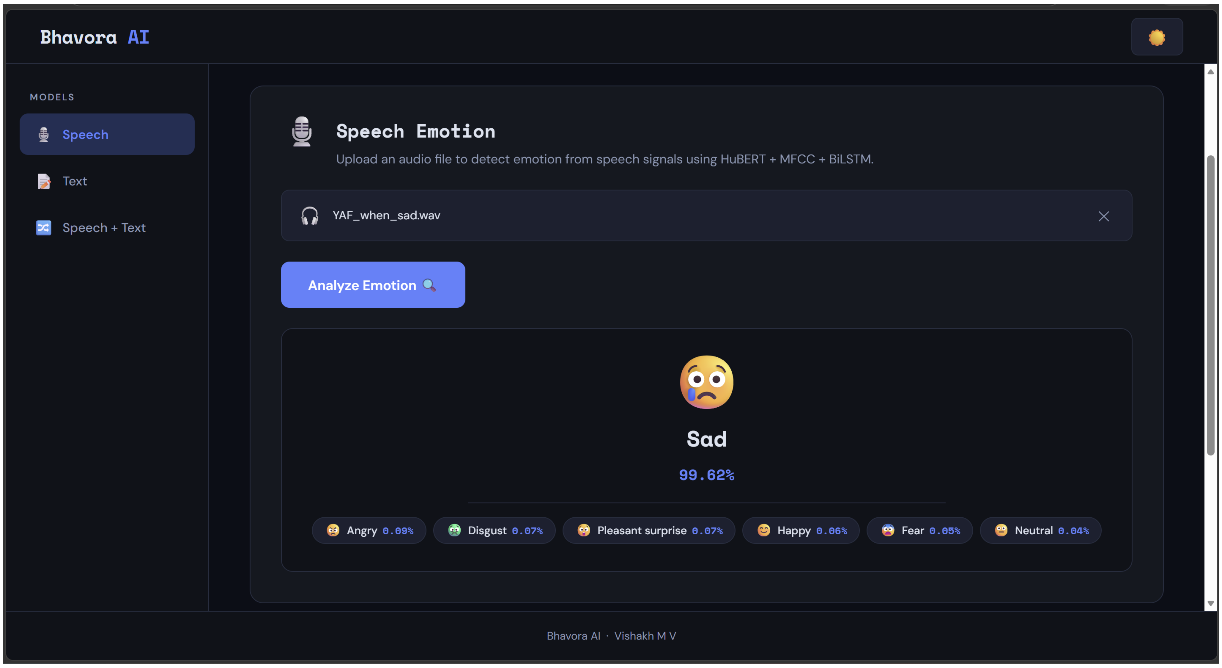


Figure A.1: Prediction result visualization interface in Dark Mode.

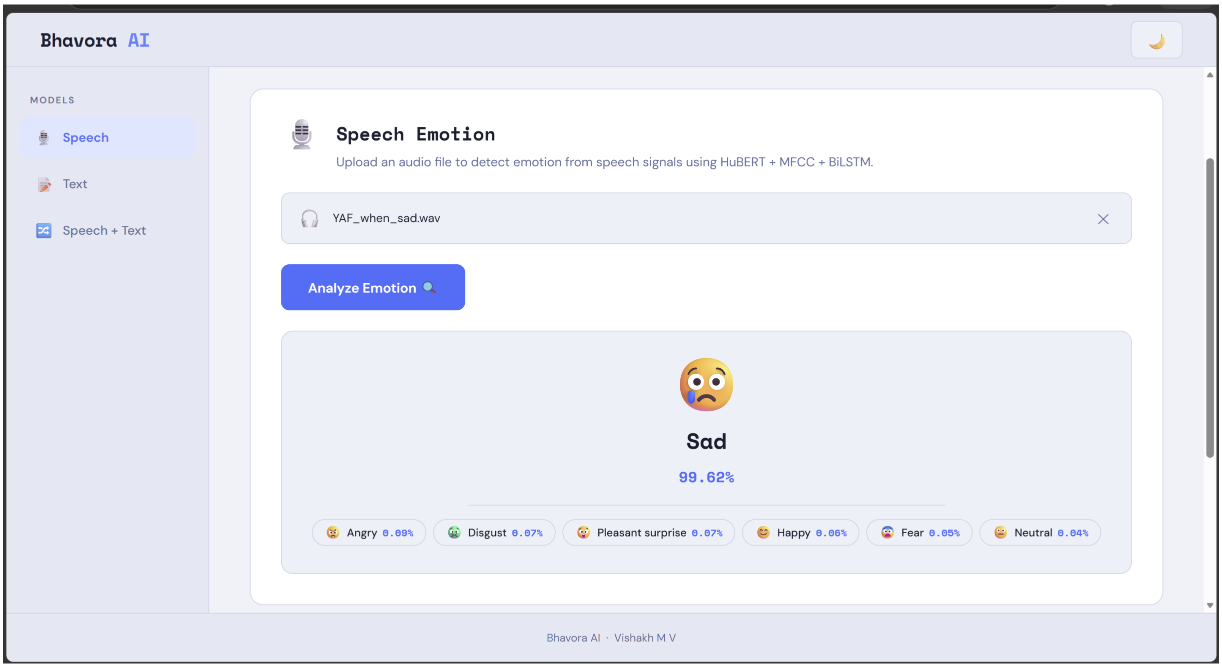


Figure A.2: Prediction result visualization interface in Light Mode.

A.1 The Meaning of "Bhavora"

The name is inspired by the Sanskrit word *"Bhava"*, meaning emotion or feeling, and the English word *"Aura,"* representing an emotional presence or atmosphere. Together, **Bhavora** symbolizes an "Emotional Aura," reflecting the system's core capability to analyze and classify human emotions from subtle patterns in speech and text.

A.2 System Architecture

The web application is built using a lightweight client-server architecture:

- **Backend (Flask & PyTorch):** A Python Flask server acts as the local backend. Upon initialization, the server preloads the locally trained weights (`.pth`) for the Speech-Only, Text-Only, and Multimodal Fusion models directly into memory. It handles local HTTP POST requests, performs real-time audio trimming and MFCC extraction via `librosa`, and tokenizes text using Hugging Face's `DistilBertTokenizer`.

- **Frontend (HTML/CSS/JavaScript):** A responsive, component-based user interface was built using vanilla HTML, CSS, and JavaScript. It communicates with the local backend asynchronously using standard HTTP requests, ensuring the page does not reload during model inference.

A.3 Features and Capabilities

The interactive dashboard features three dedicated prediction panels matching the project's deep learning pipelines:

1. **Speech Emotion:** Accepts `.wav` or `.flac` audio uploads and predicts emotion using the HuBERT + MFCC + BiLSTM backend.

2. **Text Emotion:** Accepts user-typed text to analyze semantic emotion using the DistilBERT backend.

3. **Multimodal Fusion:** Accepts both an audio file and its corresponding text transcript, passing both to the late-fusion architecture for prediction.

Additionally, the user interface features real-time probability distribution visualization and a dynamic Dark/Light theme toggle for enhanced accessibility.

A.4 Application Demonstration

The local deployment successfully handles end-to-end multimodal inference in real-time. A recorded demonstration of the working application, showing predictions across all three modalities, is available at:
https://drive.google.com/file/d/1S3PBq-C0EeLMSWoC8bE-xJpc2mXzv23d/view?usp=drive$_{l}ink$.

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